CHAPTER 6
MOMENTS

1.5.5 Turning effect of forces

1 Describe the moment of a force as a measure of its turning effect and give everyday examples

2 Define the moment of a force as moment = force × perpendicular distance from the pivot; recall and use this equation

3 State and use the principle of moments for an object in equilibrium

4 Describe an experiment to verify the principle of moments

1.5.6 Centre of gravity

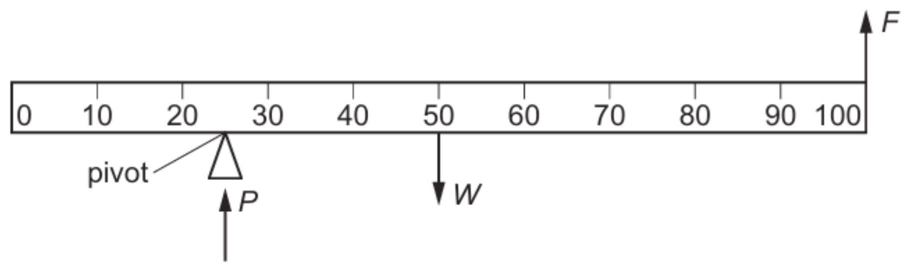
1 State what is meant by centre of gravity

2 Describe how to determine the position of the centre of gravity of a plane lamina using a plumb line

3 Describe, qualitatively, the effect of the position of the centre of gravity on the stability of simple objects

1. O/N/23/11/Q7

A uniform metre rule of weight W is pivoted at the 25 cm mark and held horizontal by a force F applied upwards at the 100 cm mark. The rule is supported by a vertical force P acting at the pivot.

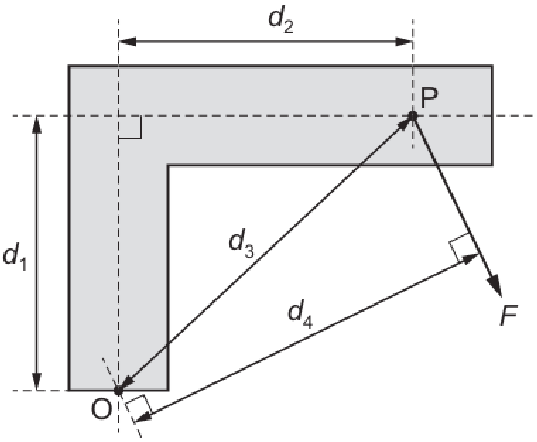


What is the magnitude of P?

A $\frac{1}{3}W$ **B** $\frac{1}{2}W$ **C** $\frac{2}{3}W$ **D** $\frac{4}{3}W$

2. O/N/23//11/Q6

A force F causes a moment about point O on an L-shaped bar. The force F acts at point P.

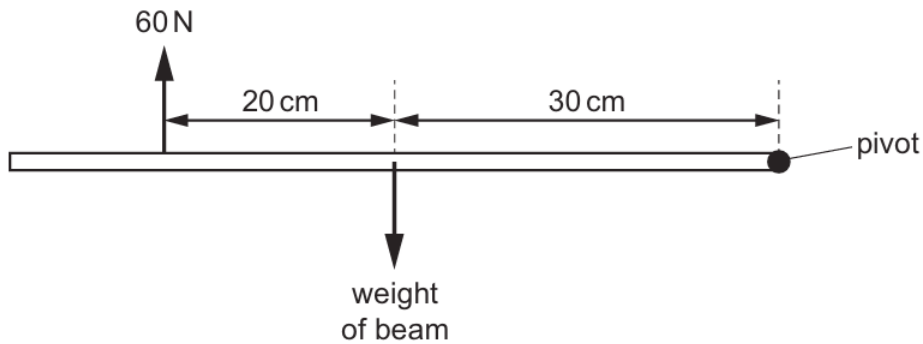


What is the moment of F about O?

A Fd_1 **B** Fd_2 **C** Fd_3 **D** Fd_4

3. M/J/23/11/Q9

A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.

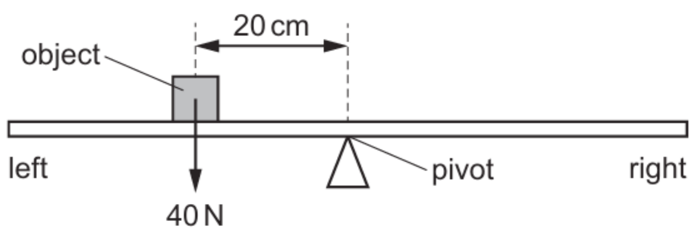


What is the weight of the beam?

A 36 N **B** 40 N **C** 90 N **D** 100 N

4. M/J/23/12/Q11

A uniform beam is pivoted at its midpoint. An object weighing 40 N is placed on the beam 20 cm to the left of the pivot as shown.



Which force and position balances the system?

A 20 N acting downwards, 40 cm to the right of the pivot

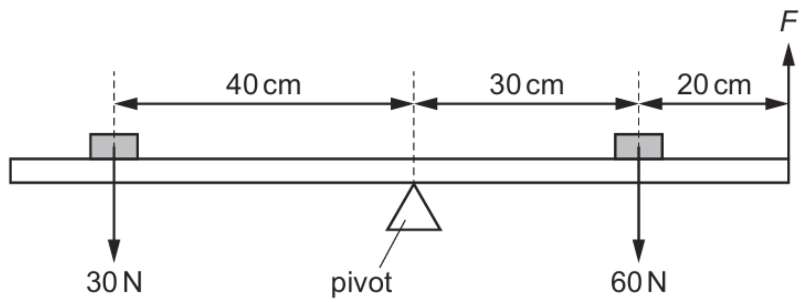
B 20 N acting upwards, 40 cm to the right of the pivot

C 50 N acting downwards, 10 cm to the left of the pivot

D 50 N acting upwards, 10 cm to the left of the pivot

5. O/N/22/11/Q11

A uniform beam is pivoted at its centre. Two weights are placed on the beam in the positions shown and the beam is balanced by an upward force *F*.



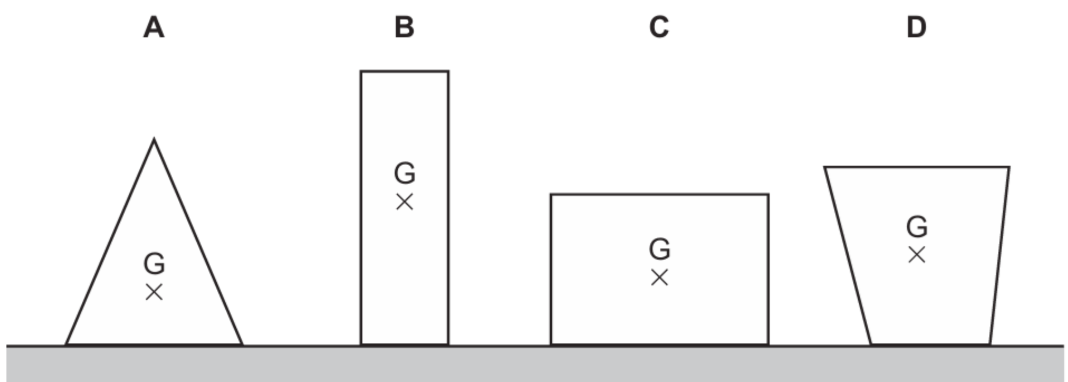
What is the size of *F*?

A 6.0 N **B** 12 N **C** 30 N **D** 60 N

6. O/N/22/12/Q13

Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

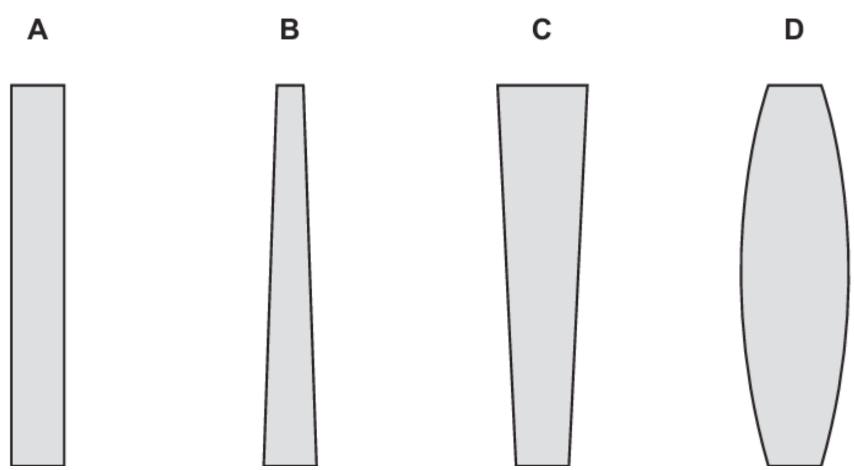
Which object is the least stable?



7. M/J/22/11/Q6

Four glass objects have square bases of equal area.

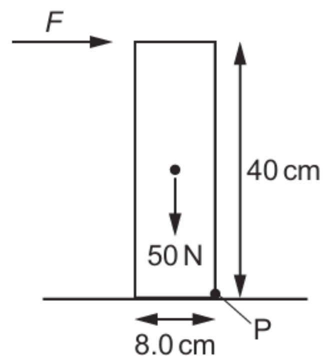
Which object is the least stable?



8. M/J/22/12/Q7

The diagram shows a uniform solid rectangular block of weight 50 N that is pivoted about point P.

The height of the block is 40 cm. The base of the block is 8.0 cm wide.

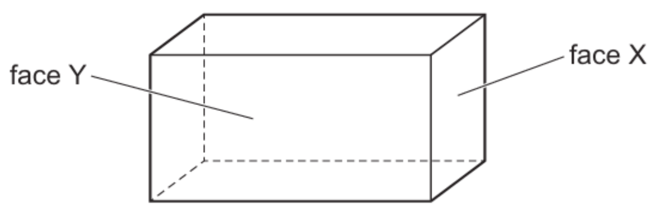


Which horizontal force *F* just makes the block start to rotate about P?

A 2.5 N **B** 5.0 N **C** 10 N **D** 160 N

9. M/J/22/12/Q8

The centre of mass of a solid rectangular block is at its centre. A small heavy weight is available.



In which arrangement is the centre of mass the lowest?

A with face X on a table

B with face Y on a table

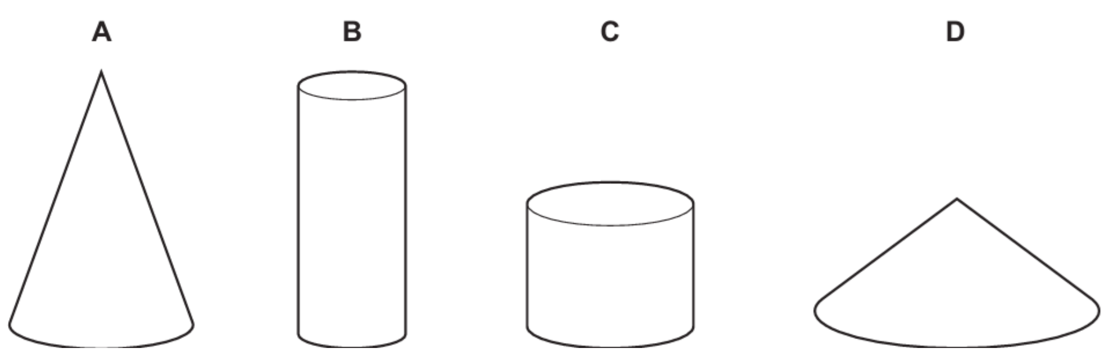
C with face X on a table and the heavy weight attached centrally on top of the block

D with face Y on a table and the heavy weight attached centrally on top of the block

10. O/N/21/12/Q8

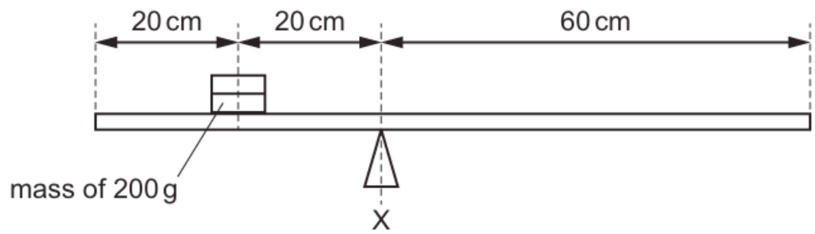
Four solid objects are made from the same material and have equal volumes.

Which object is the most stable?



11. M/J/21/11/Q8

A horizontal beam is pivoted at X. A mass of 200 g rests on the beam as shown. The centre of mass of the beam is 50 cm from the right-hand end of the beam.



The beam is balanced.

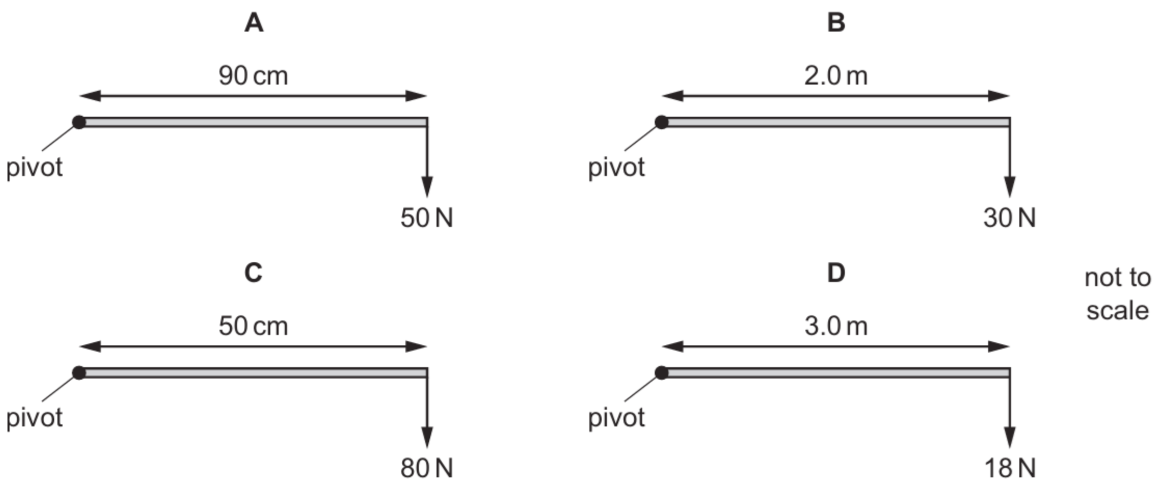
What is the mass of the beam?

A 80 g **B** 100 g **C** 400 g **D** 800 g

12. M/J/21/12/Q8

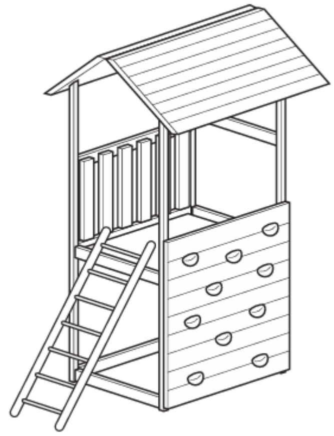
Some forces are applied at different distances from a pivot.

Which diagram shows the force that produces the largest moment about the pivot?



13. M/J/21/12/Q9

The diagram shows a children's wooden play tower.



Which change to the tower makes it more stable?

A making the tower narrower

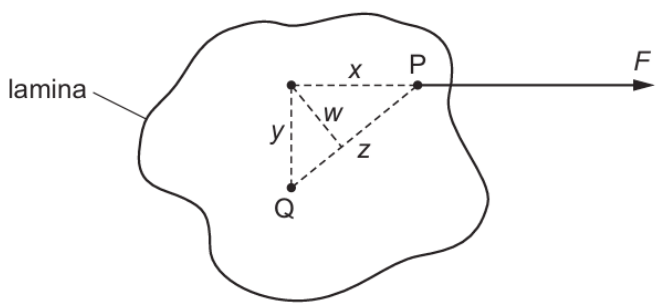
B making the tower taller

C lowering the centre of mass

D raising the centre of mass

14. O/N/20/11/Q10

A length of thread is attached to a lamina at point P, as shown in the diagram.



The lamina is free to rotate about point Q.

The tension in the thread is F.

What is the moment of F about Q?

A Fw **B** Fx **C** Fy **D** Fz

15. M/J/20/12/Q11

Which single item can be used to find the centre of mass of a plane lamina of irregular shape?

A a balance

B a measuring tape

C a micrometer

D a vertical pin

16. O/N/19/11/Q11

What is an expression for the moment of a force about a pivot?

A distance the force moves from the pivot ÷ force

B force × distance the force moves from the pivot

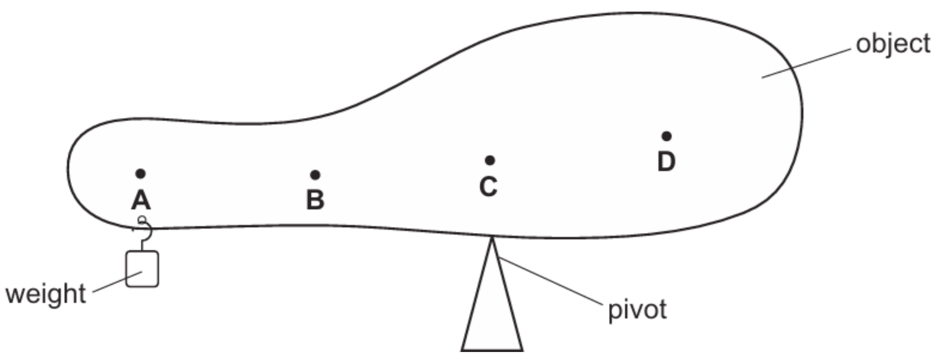
C force × perpendicular distance of the force from the pivot

D force ÷ perpendicular distance of the force from the pivot

17. O/N/19/11/Q12

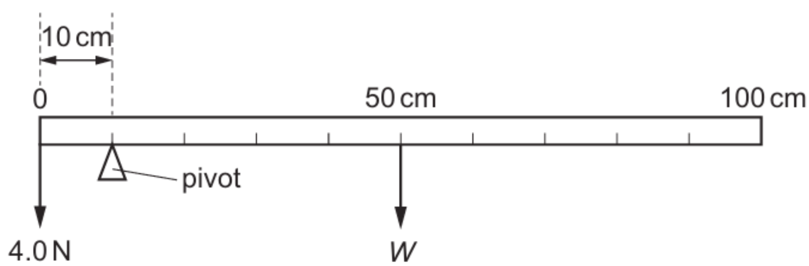
A student balances a non-uniform object on a pivot. To do this, a weight is suspended near the left-hand end of the object, as shown.

Where is the centre of mass of the object?



18. M/J/19/11/Q9

A uniform metre rule is balanced by a 4.0 N weight as shown.



What is the weight W of the metre rule?

A 1.0 N **B** 4.0 N **C** 16 N **D** 40 N

19. M/J/19/11/Q10

Which statement about centre of mass is correct?

A Objects with a centre of mass at the same height are less stable when the base is larger.

B Objects with a centre of mass at the same height are more stable when the base is larger.

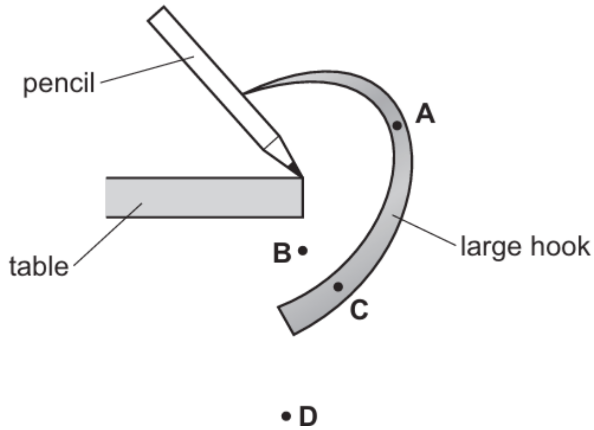
C Objects with higher centres of mass and smaller bases are more stable.

D Objects with identical bases are more stable when the centre of mass is higher.

20. M/J/19/12/Q11

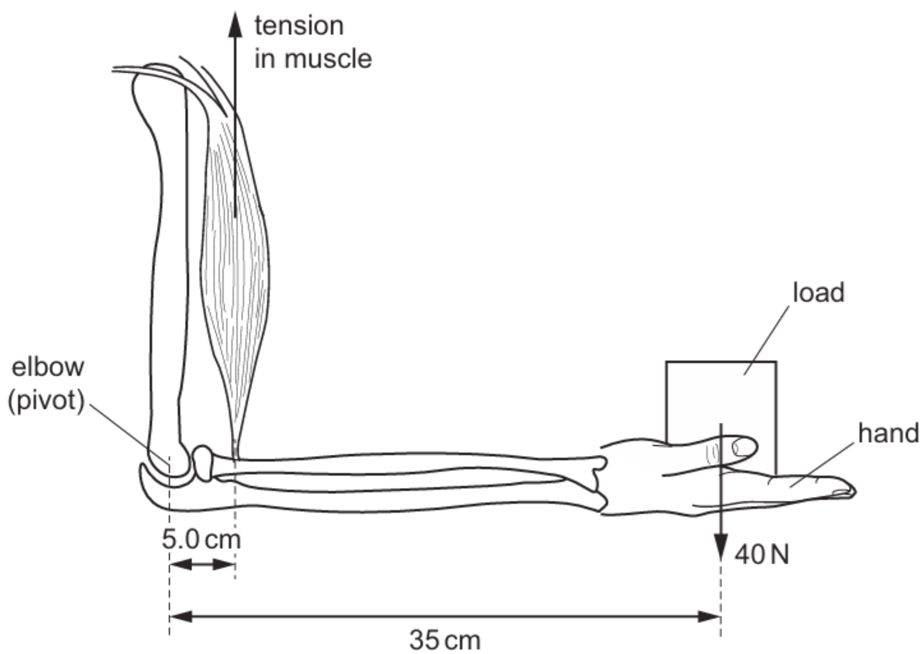
A large hook sticks into a pencil. The hook and pencil are balanced on the edge of a table, as shown.

Where is the centre of mass of the hook and pencil?



21. M/J/18/11/Q9

The diagram shows a muscle and bones in a person's arm. The hand holds a load of weight 40 N. The elbow acts as a pivot and the tension in the muscle keeps the lower part of the arm horizontal.



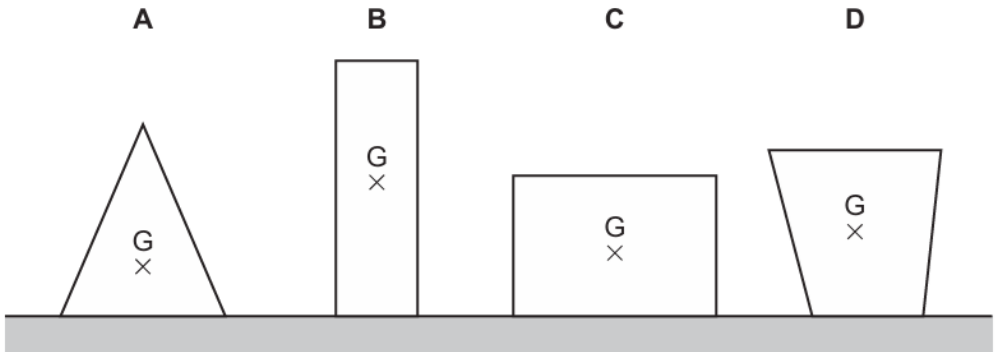
What is the tension in the muscle due to the load?

A 200 N **B** 240 N **C** 280 N **D** 1400 N

22. M/J/18/11/Q10

Four objects of equal mass rest on a table. The centre of mass of each object is labelled G.

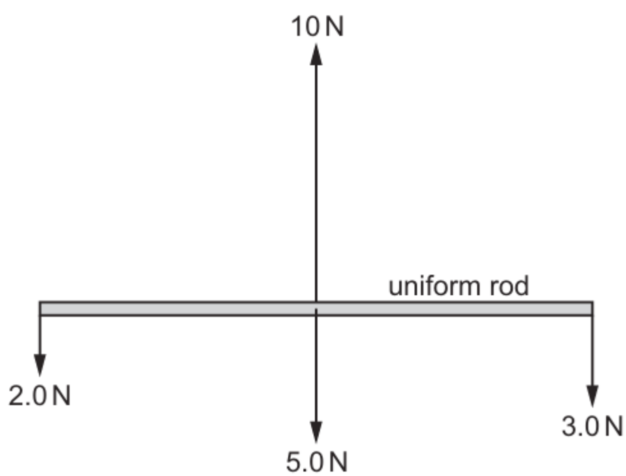
Which object is the least stable?



23. M/J/18/12/Q7

A uniform rod of weight 5.0 N is held initially at rest.

The diagram shows the forces acting on the rod when it is released.



What happens to the rod when it is released?

A It does not move.

B It moves to the right.

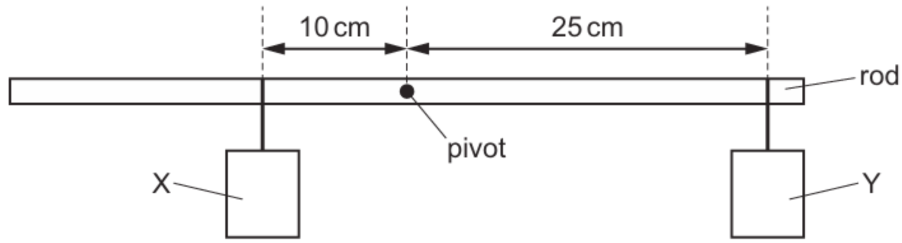
C It moves upwards.

D It starts to rotate.

24. M/J/18/12/Q10

Two objects X and Y are suspended from a uniform rod, pivoted at its centre.

The rod is in equilibrium.



Which statement about X and Y is correct?

A The mass of X is 0.4 times the mass of Y.

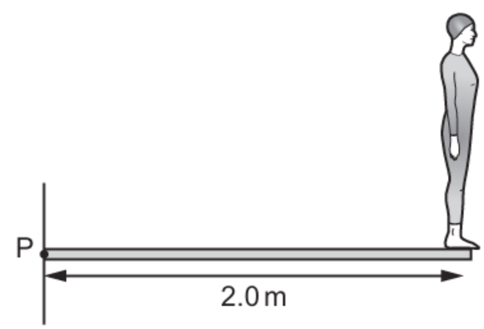
B The mass of X is 2.5 times the mass of Y.

C The mass of X is 3.5 times the mass of Y.

D The mass of X is equal to the mass of Y.

25. O/N/17/12/Q7

A diver of weight 500 N stands at the end of a springboard that is 2.0 m long and is fixed at point P.



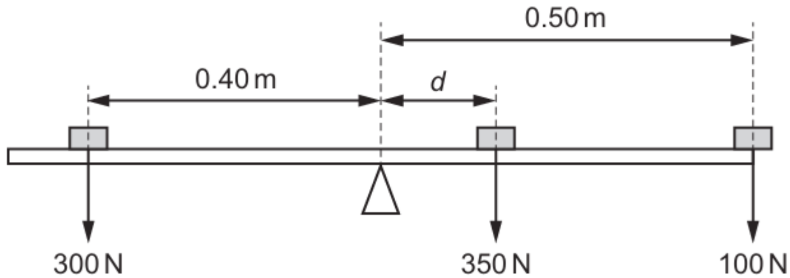
The springboard has a weight of 500 N. The centre of mass of the springboard is in the centre of the board.

What is the total moment about point P of the diver and the board?

A 500 N m **B** 750 N m **C** 1000 N m **D** 1500 N m

26. M/J/17/11/Q12

A uniform beam is pivoted at its centre. The beam is balanced by three weights in the positions shown.

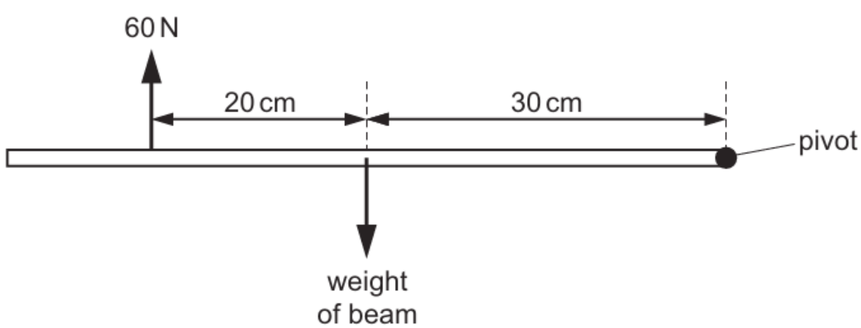


What is the length *d*?

A 0.020 m **B** 0.050 m **C** 0.20 m **D** 0.48 m

27. M/J/16/11/Q9

A uniform horizontal beam, pivoted at its right-hand end, is in equilibrium. A force of 60 N acts vertically upwards on the beam as shown.



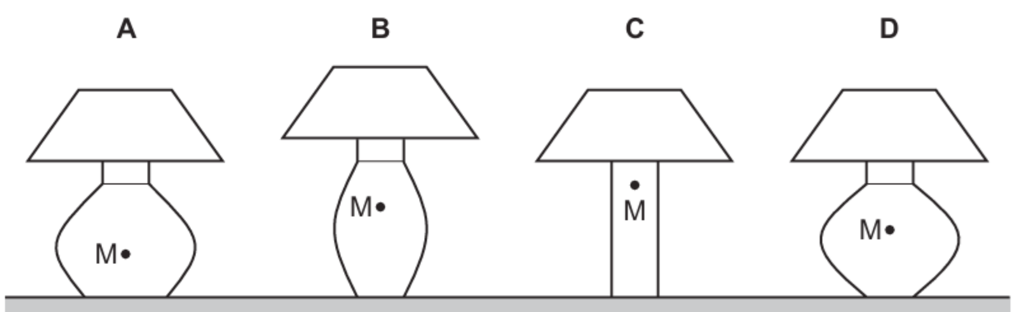
What is the weight of the beam?

A 36 N **B** 40 N **C** 90 N **D** 100 N

28. M/J/16/11/Q10

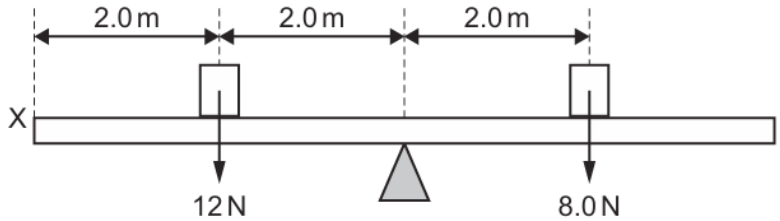
Four table lamps are shown along with the position M of the centre of mass in each case.

Which lamp is the most stable?



29. M/J/16/12/Q6

A uniform plank is pivoted at its mid-point. Two weights are added to the plank, one weight on each side of the pivot in the positions shown.



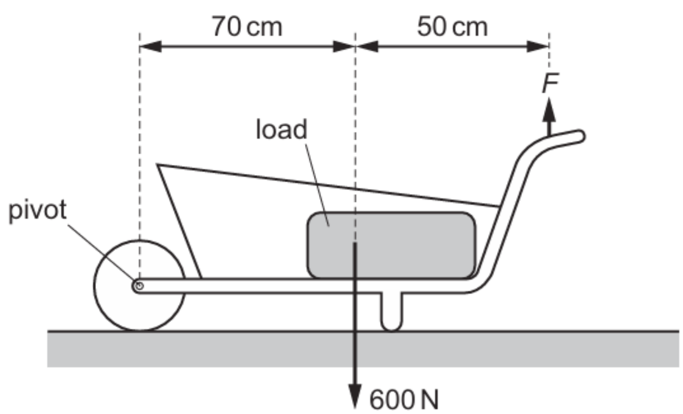
A vertical force is applied at point X to balance the plank.

What is the size and direction of this force?

	size / N	direction
A	2.0	downwards
B	2.0	upwards
C	4.0	downwards
D	4.0	upwards

30. O/N/15/11/Q8

The total weight of the load and the wheelbarrow shown is 600 N.



What is the size of force F needed just to lift the loaded wheelbarrow?

A 350 N **B** 430 N **C** 600 N **D** 840 N

31. M/J/15/11/Q10

When a car turns a corner at speed, it risks toppling over. Two factors affecting the stability of a car are the height of its centre of mass and the distance between its front wheels.

Which factors make the car most stable?

	centre of mass	distance between front wheels
A	high	small
B	high	large
C	low	small
D	low	large

32. O/N/14/11/Q7

What affects the stability of an object?

A only its base area and the location of its centre of mass

B only its weight and its base area

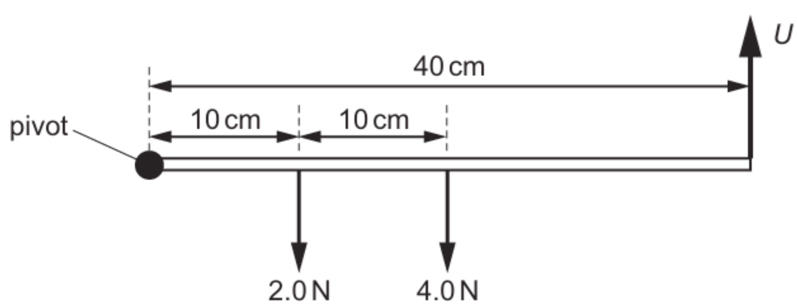
C only the location of its centre of mass

D only its weight

33. M/J/14/11/Q9

A beam of length 40 cm is pivoted at one end.

The weight of the beam is 4.0 N and acts at a point 20 cm from the pivot. A 2.0 N weight hangs 10 cm from the pivot.



An upward force *U* is needed to keep the beam horizontal.

What is the size of *U*?

A 0.5 N **B** 1.5 N **C** 2.5 N **D** 6.0 N

34. M/J/14/11/Q10

A car is designed to be stable.

To achieve good stability, where is the centre of mass of the car?

A above the front wheels

B above the rear wheels

C as high in the car as possible

D as low in the car as possible

35. M/J/14/12/Q10

A man uses clay to make a pot. He wants the pot to be as stable as possible when placed on a flat surface.

Which two features of the pot must the man consider?

A the area of the base and the height of the centre of gravity

B the density of the clay and the area of the base

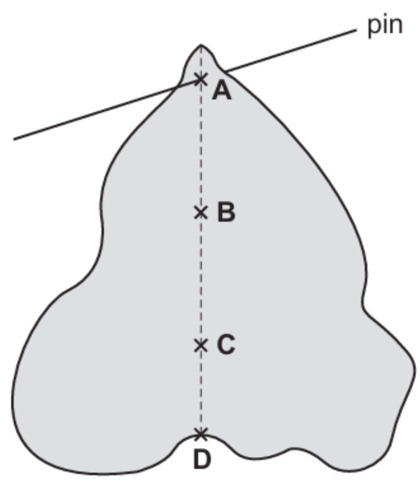
C the density of the clay and the height of the centre of gravity

D the weight and the height of the centre of gravity

36. O/N/13/11/Q7

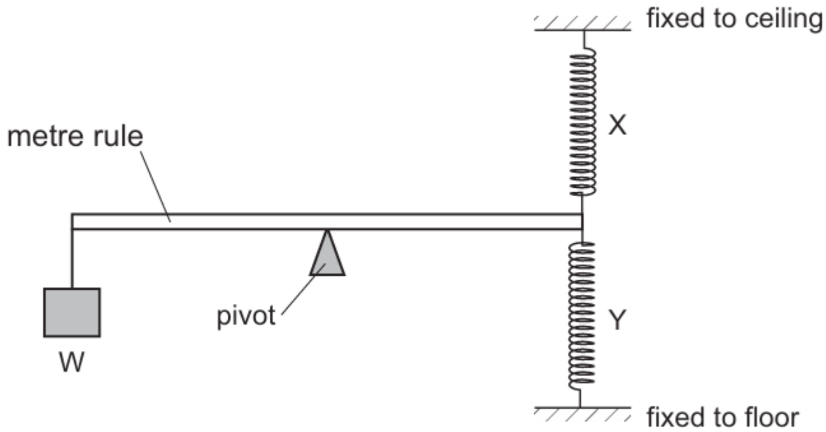
A piece of uniform card is suspended freely from a horizontal pin.

Which point is its centre of mass?



37. O/N/13/12/Q7

Two stretched springs X and Y are attached to one end of a metre rule as shown. A weight W is hung from the other end. A pivot is at the centre of the rule.



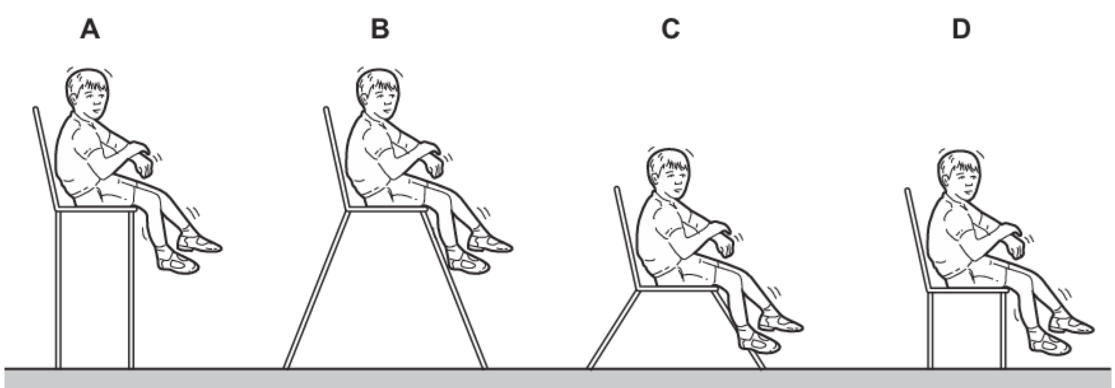
The weight W is moved towards the pivot.

How does the extension of each spring change?

	spring X	spring Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

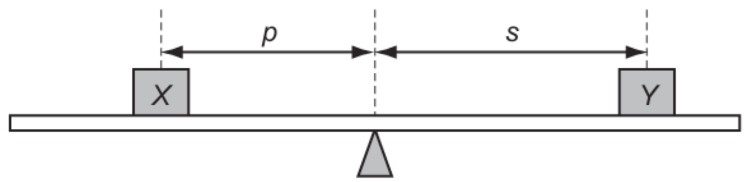
38. M/J/13/11/Q7

Which chair is the **least** stable if the child moves?



39. O/N/12/11/Q7

Masses *X* and *Y* are placed on opposite sides of the centre of a uniform metre rule, which is pivoted at its centre.

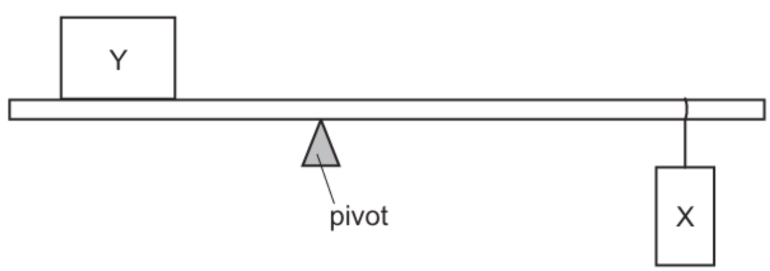


Which combination of masses and distances balances the rule?

	mass / g		distance / cm	
	X	*Y*	*p*	*s*
A	200	200	5	10
B	200	300	10	15
C	400	300	12	16
D	500	200	15	30

40. O/N/12/12/Q7

An object Y is in a fixed position on a rod. A weight X is moved and the position of a pivot is adjusted until the rod balances on the pivot, as shown.



The experiment is repeated in a region where the gravitational field strength is lower.

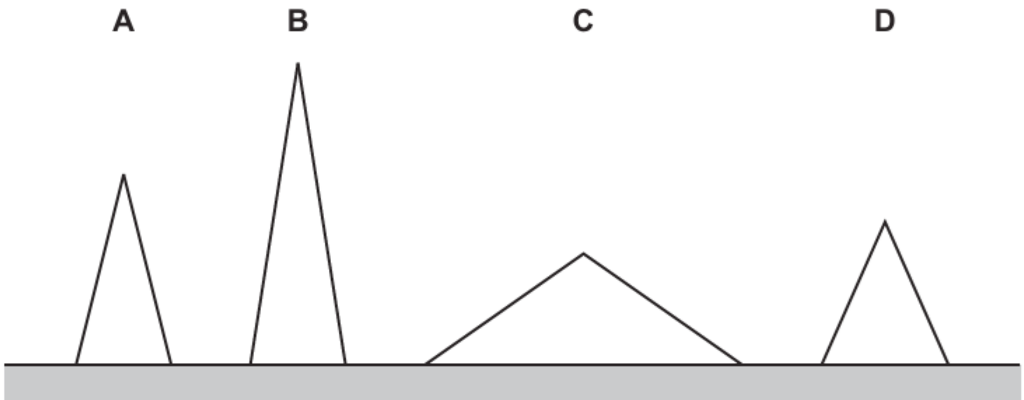
What is done to keep the rod balanced?

	pivot	X
A	move left	no movement
B	move right	move left
C	no movement	move right
D	no movement	no movement

41. O/N/12/12/Q8

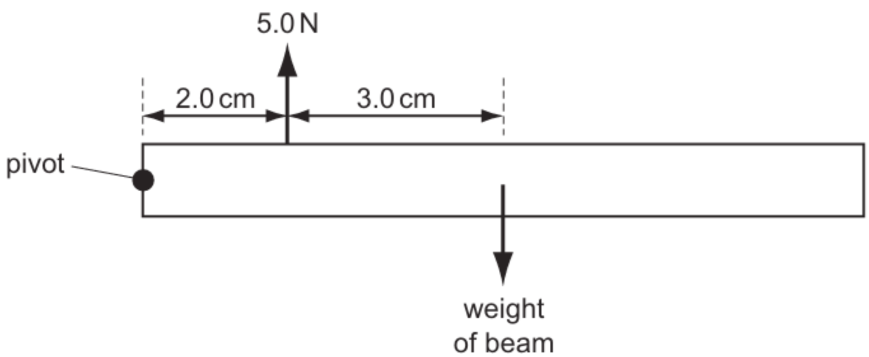
Four solid uniform cones have equal weight. They are placed on a bench as shown in the scale diagram.

Which cone is the most stable?



42. M/J/12/11/Q7

A beam pivoted at one end has a force of 5.0 N acting vertically upwards on it as shown. The beam is in equilibrium.

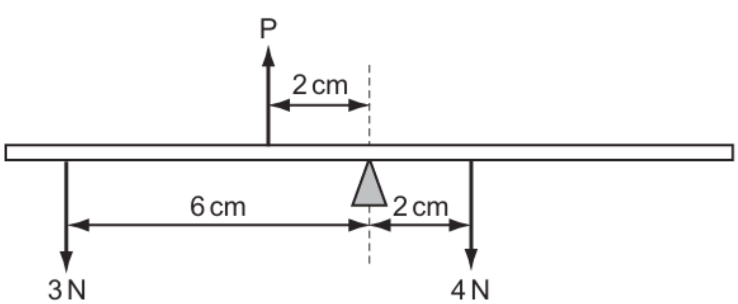


What is the weight of the beam?

A 2.0 N **B** 3.0 N **C** 3.3 N **D** 5.0 N

43. M/J/12/12/Q7

The diagram shows a uniform balanced beam, pivoted about its centre.



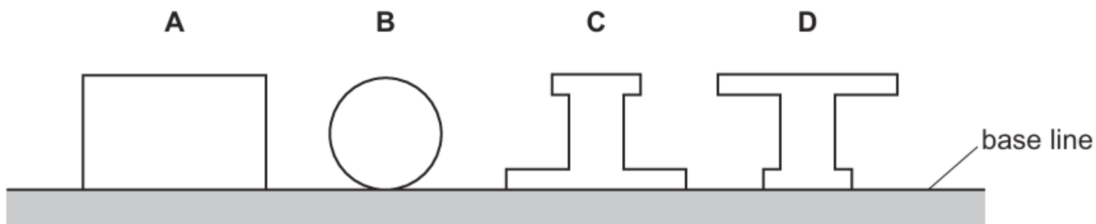
What is the value of force P?

A 5 N **B** 7 N **C** 10 N **D** 13 N

44. M/J/12/12/Q8

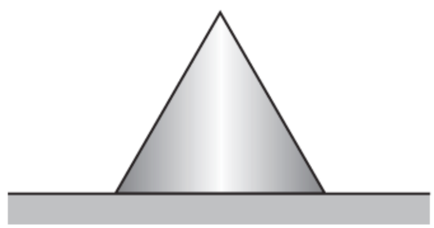
The diagram shows four shapes, cut from the same piece of card.

Which shape has its centre of mass nearest to the base line?



45. O/N/11/11/Q9

A metal cone with a circular base is placed on a flat surface.



The stability of the cone depends on

A its weight only.

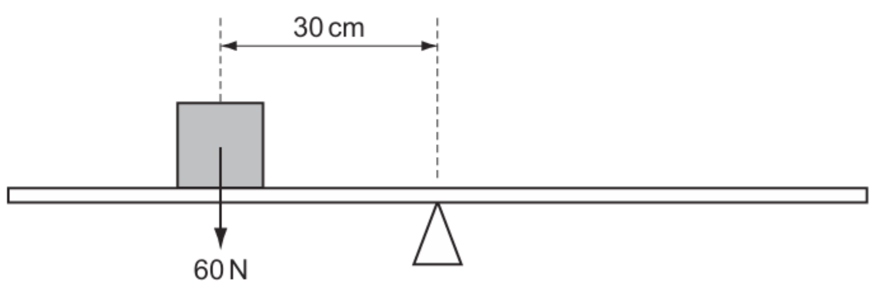
B the diameter of its base and the position of its centre of mass.

C the diameter of its base only.

D the position of its centre of mass only.

46. M/J/11/11/Q8

A uniform beam is balanced at its midpoint. An object is placed on the beam, as shown.



Which force will rebalance the beam?

A 30 N acting upwards, 60 cm to the left of the midpoint

B 30 N acting upwards, 60 cm to the right of the midpoint

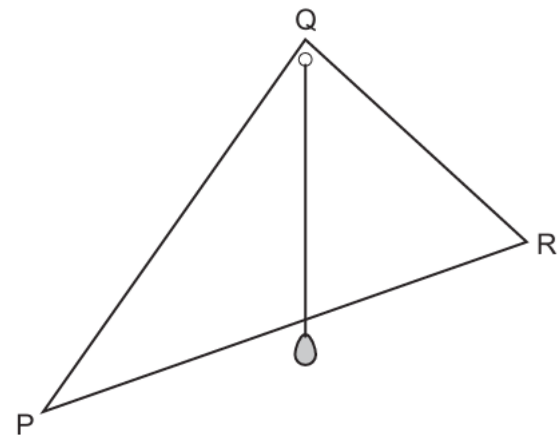
C 45 N acting downwards, 45 cm to the right of the midpoint

D 90 N acting downwards, 20 cm to the left of the midpoint

47. M/J/11/11/Q9

A student finds the centre of mass of a triangular lamina PQR.

He drills a small hole at Q. He suspends the lamina from a pin through the hole at Q so that the lamina swings freely. He then hangs a plumb-line from the pin at Q, as shown. He marks the position of the plumb-line on the lamina.



To determine the location of the centre of mass, the student then repeats the experiment but with one change.

What is the change?

A He suspends the lamina from the hole at Q, with R on the left and P on the right.

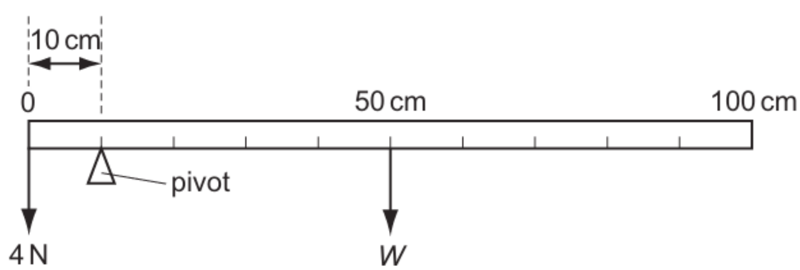
B He suspends the lamina from a pin through a hole at R.

C He uses a heavier weight on the plumb-line.

D He uses a longer plumb-line.

48. O/N/10/11/Q10

A uniform metre rule is balanced by a 4 N weight as shown in the diagram.

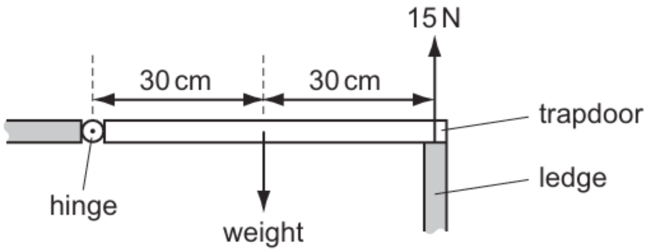


What is the weight *W* of the metre rule?

A 1 N **B** 4 N **C** 16 N **D** 40 N

49. M/J/10/11/Q8

A wooden trapdoor is hinged along one side and, when closed, is supported on the other side by a ledge.



When the trapdoor is closed, the ledge exerts an upward force of 15 N on the trapdoor. The gravitational field strength is 10 N / kg.

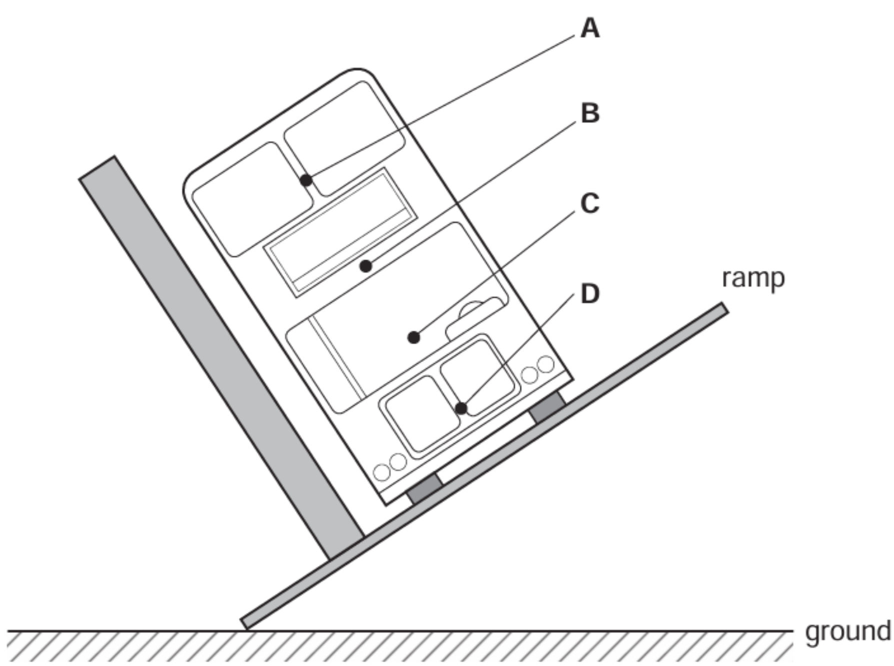
What is the mass of the trapdoor?

A 1.5 kg **B** 3.0 kg **C** 30 kg **D** 150 kg

50. O/N/09/Q8

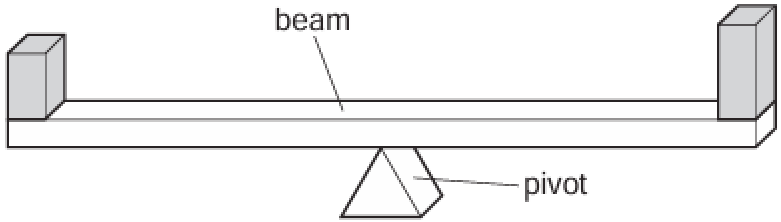
The stability of a bus is tested by tilting it on a ramp. The diagram shows a bus that is just about to topple over.

Where is the centre of mass of the bus?



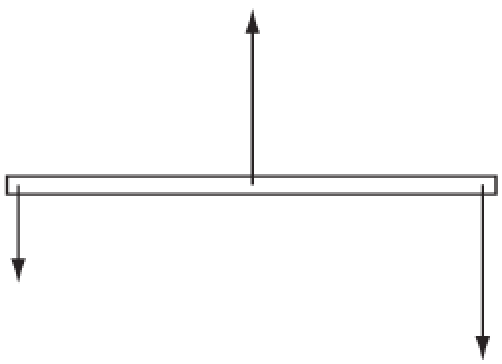
51. M/J/09/Q8

Two blocks are placed on a beam which balances on a pivot at its centre. The weight of the beam is negligible.

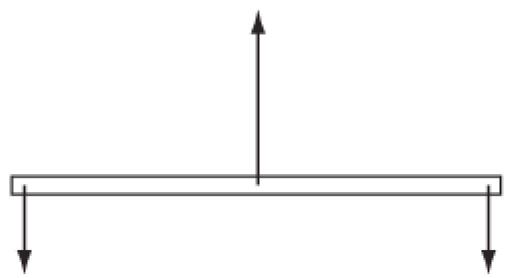


Which diagram shows the forces acting on the beam?
(The length of each arrow represents the size of a force.)

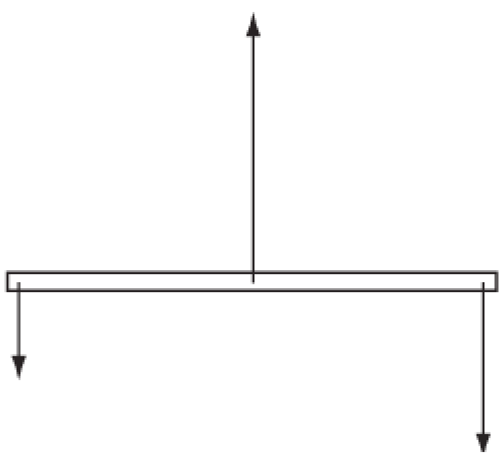
A



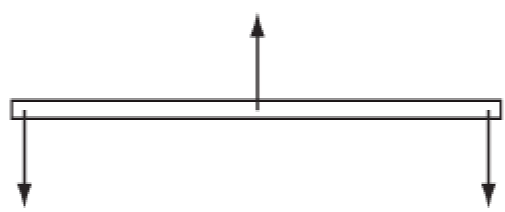
B



C

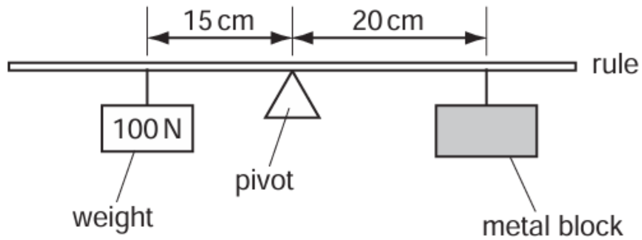


D



52. O/N/08/Q7

The diagram shows a uniform half-metre rule balanced at its mid-point.



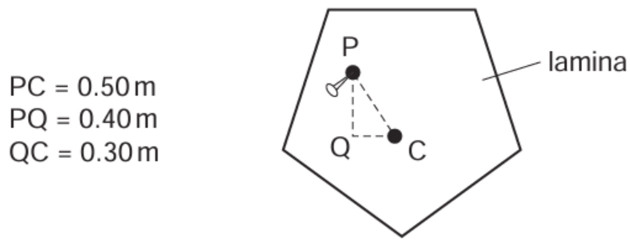
What is the weight of the metal block?

A 50 N **B** 75 N **C** 100 N **D** 150 N

53. M/J/08/Q8

A flat lamina is freely suspended from point P.

The weight of the lamina is 2.0 N and the centre of mass is at C.



The lamina is displaced to the position shown.

What is the moment that will cause the lamina to swing?

A 0.60 N m clockwise

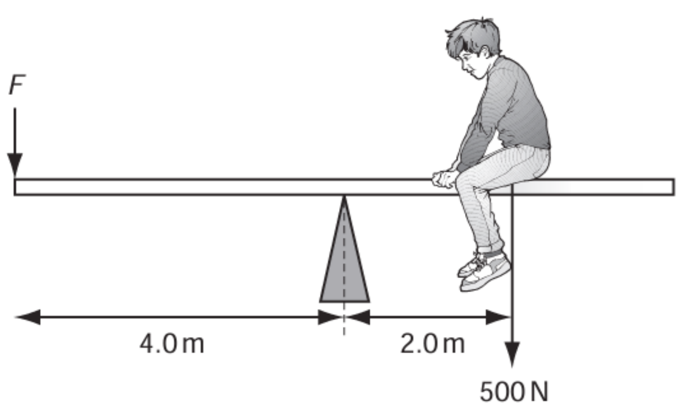
B 0.80 N m anticlockwise

C 1.0 N m clockwise

D 1.0 N m anticlockwise

54. O/N/07/Q7

The diagram shows a boy of weight 500 N sitting on a see-saw. He sits 2.0 m from the pivot.



What is the force F needed to balance the see-saw?

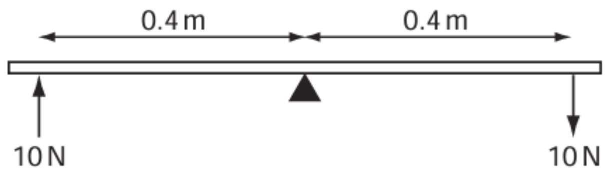
A 250 N **B** 750 N **C** 1000 N **D** 3000 N

55. M/J/07/Q7

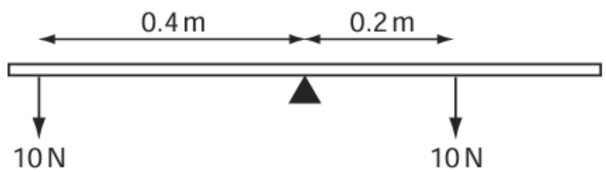
Forces are applied to a uniform beam pivoted at its centre.

Which beam is balanced?

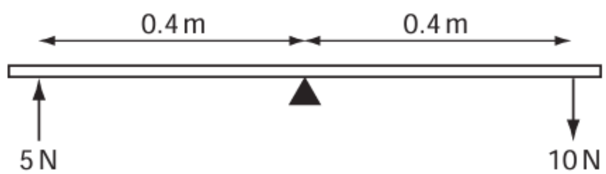
A



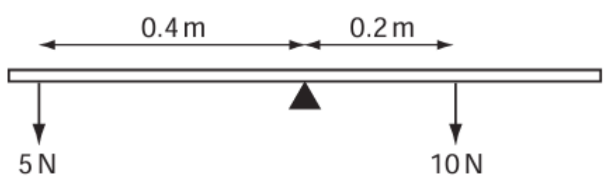
B



C



D



56. M/J/06/Q8

If a nut and bolt are difficult to undo, it may be easier to turn the nut by using a longer spanner.

This is because the longer spanner gives

A a larger turning moment.

B a smaller turning moment.

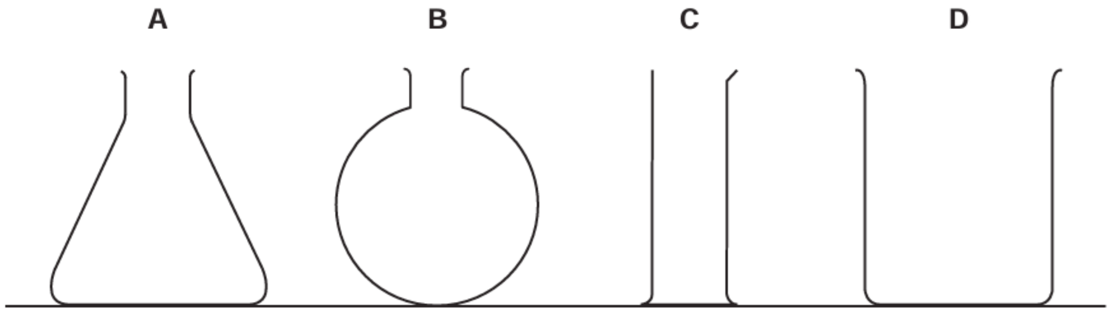
C less friction.

D more friction.

57. M/J/06/Q9

Some containers are made from thin glass.

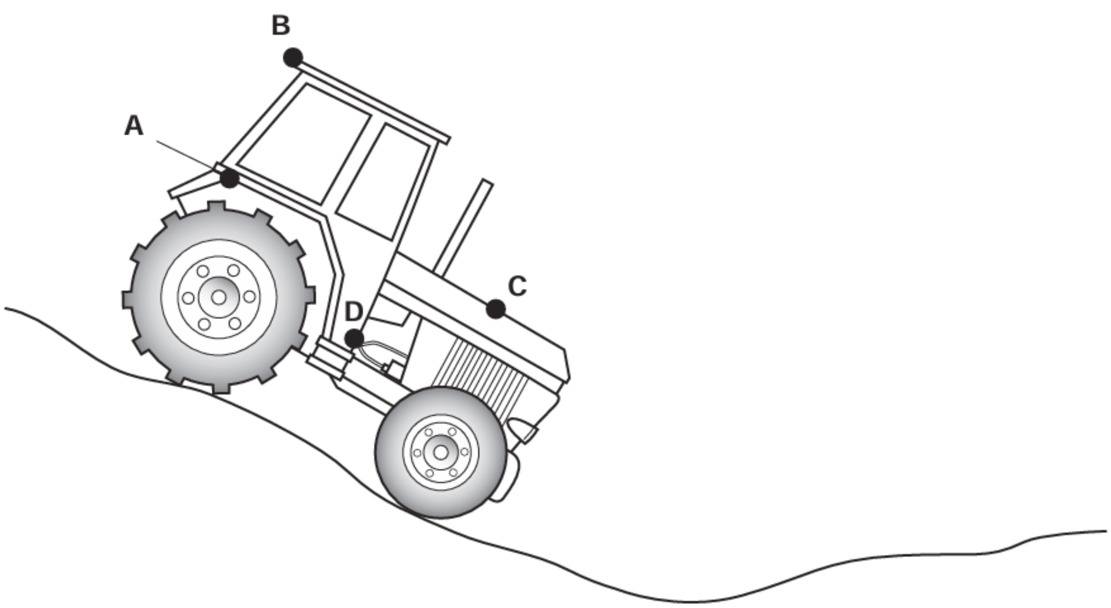
Which empty container is the **most** stable?



58. O/N/05/Q8

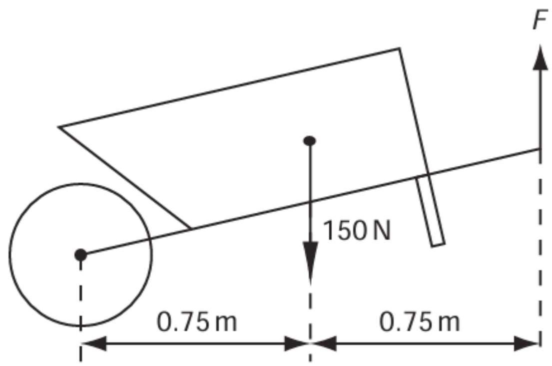
A tractor is being used on rough ground.

What is the safest position for its centre of mass?



59. M/J/05/Q6

The diagram shows a wheelbarrow and its load, which have a total weight of 150 N. This is supported by a vertical force *F* at the ends of the handles.



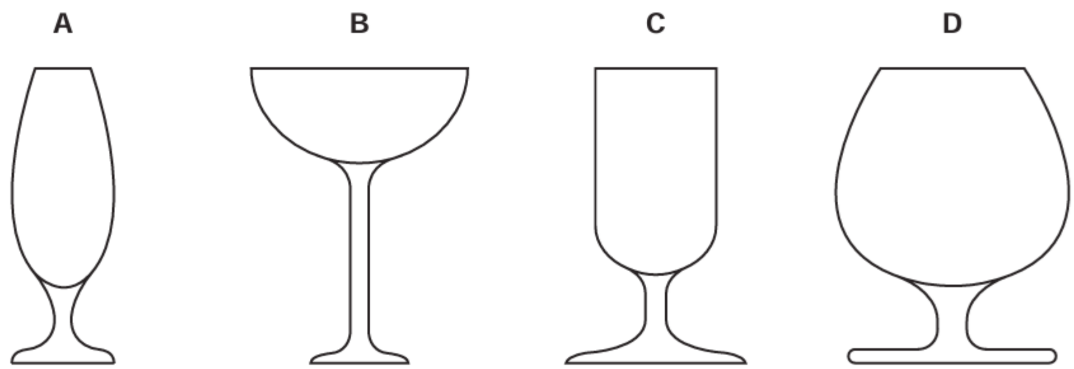
What is the value of *F*?

A 75 N **B** 150 N **C** 225 N **D** 300 N

60. M/J/05/Q7

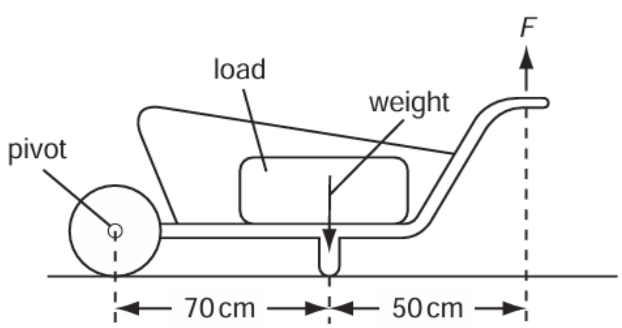
The diagrams show the cross-sections of different glasses.

Which one is the least stable when filled with a liquid?



61. O/N/04/Q8

A load is to be moved using a wheelbarrow. The total mass of the load and wheelbarrow is 60 kg. The gravitational field strength is 10 N / kg.

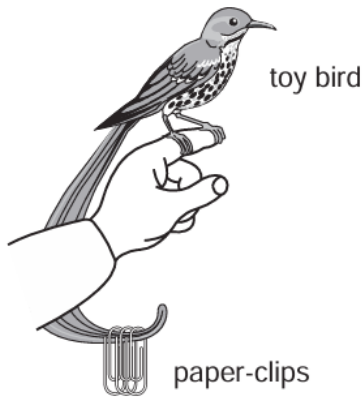


What is the size of force *F* needed just to lift the loaded wheelbarrow?

A 350 N **B** 430 N **C** 600 N **D** 840 N

62. O/N/04/Q9

A girl uses paper-clips to balance a toy bird on her finger as shown.



What is the effect of the paper-clips?

A They help to raise the centre of mass above her finger.

B They help to raise the centre of mass to her finger.

C They help to lower the centre of mass below her finger.

D They do not affect the centre of mass but increase the weight.

ANSWERS

1. C
2. D
3. D
4. A
5. B
6. B
7. C
8. B
9. B
10. D
11. C
12. B
13. C
14. C
15. D
16. C
17. D
18. A
19. B
20. B
21. C
22. B
23. D
24. B
25. D
26. C
27. D
28. A
29. B
30. A
31. D
32. A
33. C
34. D
35. A
36. C
37. C
38. A
39. C
40. D
41. C
42. A
43. A
44. C
45. B
46. A
47. B
48. A
49. B
50. C
51. B
52. B
53. A
54. A
55. D
56. A
57. A
58. D
59. A
60. B
61. A
62. C

1. O/N/23/22/Q3

A plane, irregular lamina has a mass of 50 g. It hangs from a nail that passes through a small hole H near to the edge of the lamina.

The nail acts as a pivot and the lamina can swing about it. The lamina is held in the position shown in Fig. 3.1, a small distance above a horizontal bench.

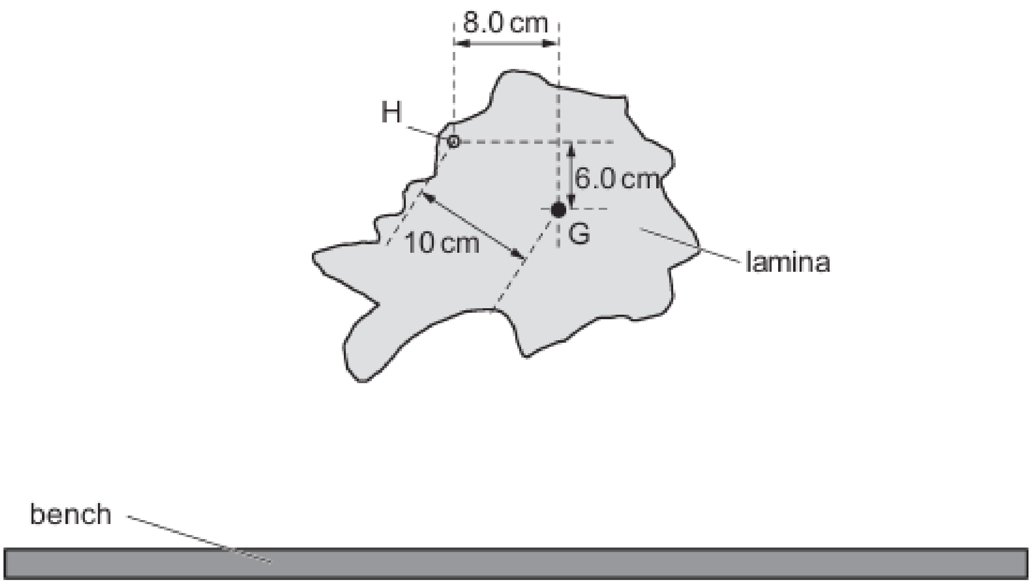


Fig. 3.1 (not to scale)

The centre of gravity of the lamina is at G.

(a) The weight of the lamina is a force that acts downwards.

(i) Explain why the lamina experiences this downward force.

....

.... [1]

(ii) Calculate the weight of the lamina on Earth.

weight = ... N [2]

(b) The weight produces a moment about the pivot through H.

(i) Using Fig. 3.1, calculate the moment of the weight about the pivot.

moment = ... N m [2]

(ii) The lamina is now released from the position shown in Fig. 3.1.

Describe what happens to the lamina from the time it is released until it stops moving.

....

....

....

.... [3]

[Total: 8]

2. M/J/22/21/Q1

Fig. 1.1 shows a model of the human arm. The rubber band represents the muscle that moves part of the arm XY up.

A mass is suspended from XY, as shown in Fig. 1.2. The weight of section XY is negligible and the model is at rest.

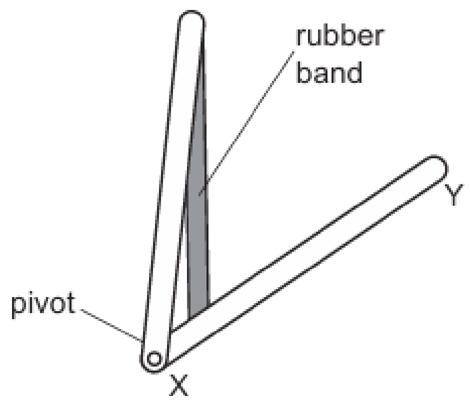


Fig. 1.1

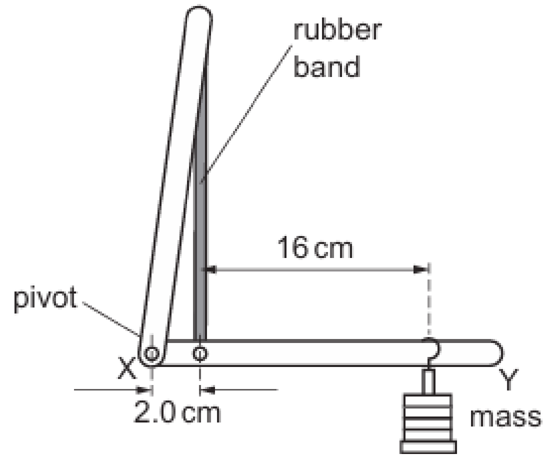


Fig. 1.2 (not to scale)

(a) **(i)** State **two** ways in which the dimensions of the rubber band change as the mass is added to section XY.

....

.... [2]

(ii) State the form of energy stored in the stretched rubber band.

.... [1]

(b) **(i)** State the principle of moments.

....

....

....

.... [2]

(ii) Explain why the force that the rubber band exerts on section XY is larger than the weight of the mass.

....

....

.... [1]

(iii) The mass suspended from section XY in Fig. 1.2 has a weight of 4.0 N.

Calculate the force that the rubber band exerts on section XY.

force = ... [2]

(iv) Explain how your answer to **(b)(iii)** is different if the weight of section XY is **not** negligible.

....

....

.... [1]

[Total: 9]

3. O/N/21/21/Q1

Fig. 1.1 shows a wooden bench of weight 2000 N.

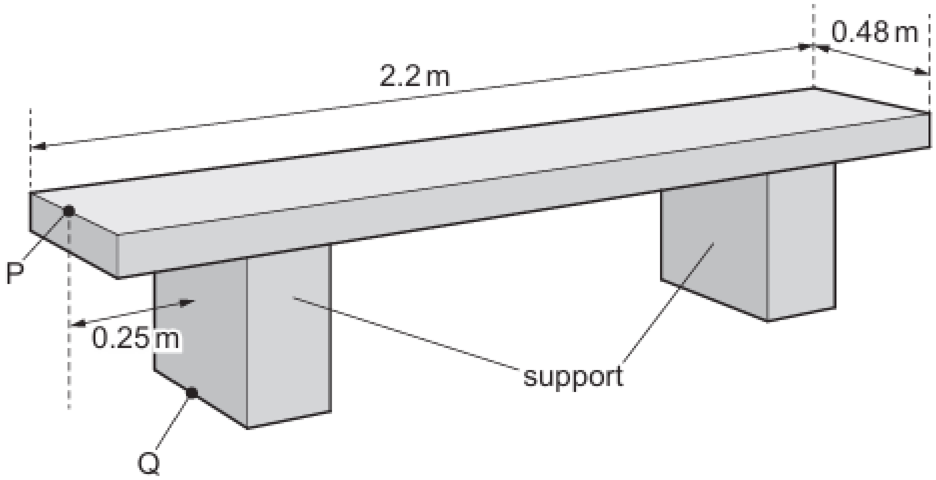


Fig. 1.1

(a) Each of the two supports has an area of $0.040\,m^2$ in contact with the ground.

Calculate the pressure on the ground due to the bench.

pressure = ... [2]

(b) The centre of mass of the bench is 1.1 m from the left-hand end of the bench and 0.24 m from the front.

(i) Suggest **one** reason why the centre of mass is in this position.

....

....

.... [2]

(ii) There is a force exerted vertically downwards from the point P shown in Fig. 1.1.

Calculate the maximum force that can be exerted vertically downwards at P without the bench rotating about the point Q shown in Fig. 1.1.

maximum force = ... [3]

[Total: 7]

4. O/N/21/22/Q8

The turning effect of a force is measured by its moment.

(a) Fig. 8.1 shows a force *F* acting on an object at point P. The object is free to rotate about an axis at X that is perpendicular to the page.

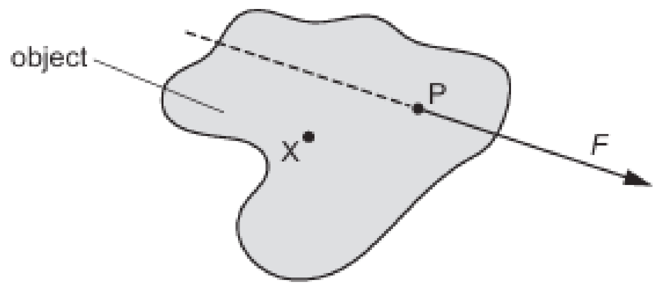


Fig. 8.1

Write down an expression for the moment of *F* about the axis at X. Draw on Fig. 8.1 to show what is meant by any other term used in your expression.

....

.... [2]

(b) **(i)** State the principle of moments.

....

....

.... [2]

(ii) Describe an experiment to verify the principle of moments. Include a diagram to help the description.

....

....

....

....

....

.... [3]

(c) A worker carries a ladder on his shoulder. His shoulder acts as a pivot. Fig. 8.2 shows that the ladder is horizontal.

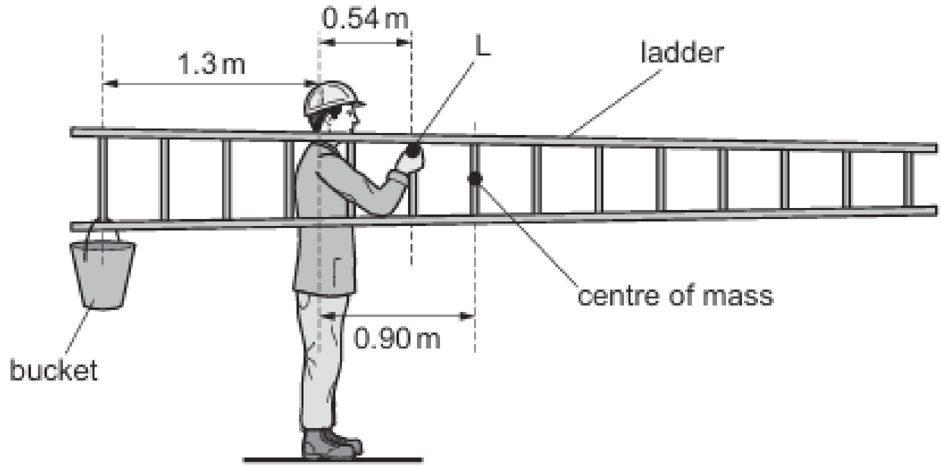


Fig. 8.2

The ladder is wider at one end than at the other end. The mass of the ladder is 8.0 kg.

(i) The gravitational field strength is 10 N / kg.

Calculate the weight of the ladder.

weight = ... [1]

(ii) State what is meant by *centre of mass*.

....

.... [1]

(iii) The centre of mass of the ladder is not halfway along its length.

State what this shows about the ladder.

....

.... [1]

(iv) The centre of mass of the ladder is a horizontal distance of 0.90 m from the worker's shoulder.

Calculate the moment about the worker's shoulder of the weight of the ladder.

moment = ... [2]

(v) A bucket of weight 87 N is suspended from the ladder at a horizontal distance of 1.3 m from the worker's shoulder.

The worker keeps the ladder horizontal by exerting a vertical force at point L. L is a horizontal distance of 0.54 m from his shoulder.

Determine the size and direction of the force exerted at L.

size of force = ...

direction = ...
[3]

[Total: 15]

5. M/J/21/22/Q3

Fig. 3.1 shows a small brick hanging from a newton meter.

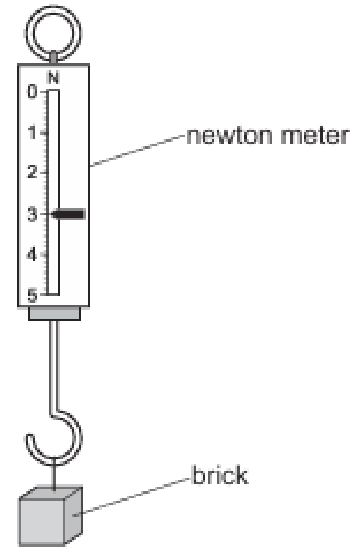


Fig. 3.1

The reading on the newton meter is 3.0 N.

(a) Describe how the reading on the newton meter is used to find the mass of the brick.

....

.... [1]

(b) The same brick and newton meter are used in the apparatus shown in Fig. 3.2. The meter rule is pivoted at its centre and is balanced. The reading on the newton meter is not shown.

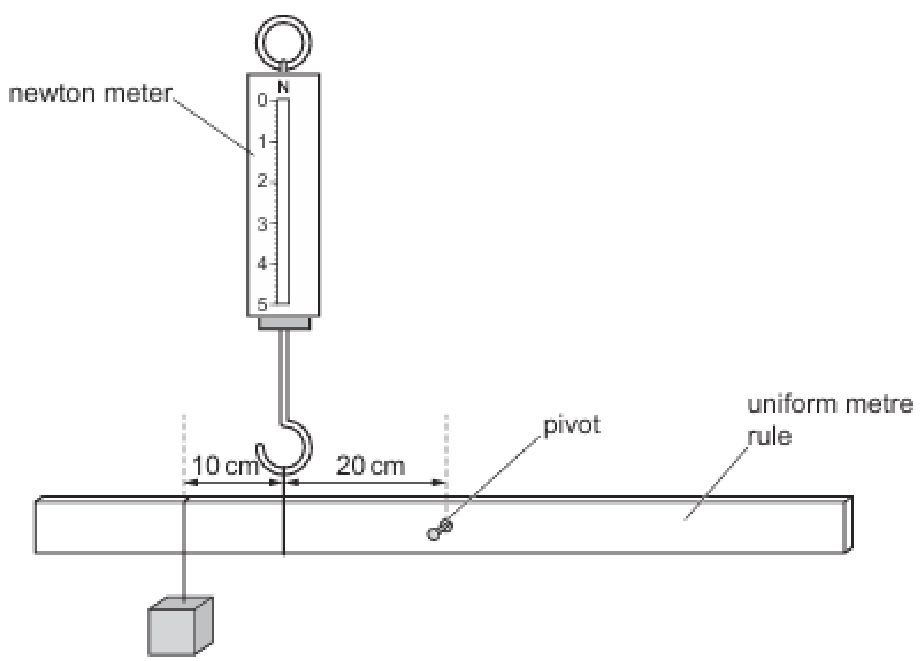


Fig. 3.2

(i) State the *principle of moments* for a body in equilibrium.

....

....

.... [1]

(ii) Determine the reading on the newton meter shown in Fig. 3.2.

reading = ... [2]

(c) A beaker of water is placed so that the brick is partly submerged in the water, as shown in Fig. 3.3. The apparatus is adjusted to keep the rule horizontal.

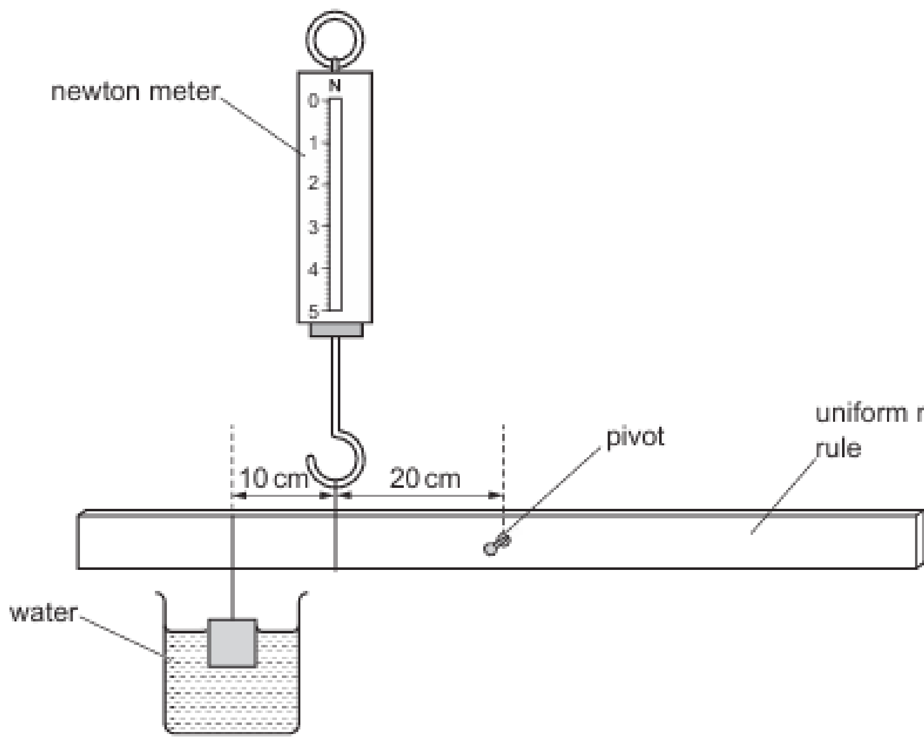


Fig. 3.3

Suggest why the reading on the newton meter is less than your answer in **(b)(ii)**.

....

....

.... [2]

[Total: 6]

6. O/N/20/21/Q3

Fig. 3.1 shows a door and an automatic door-closer viewed from above.

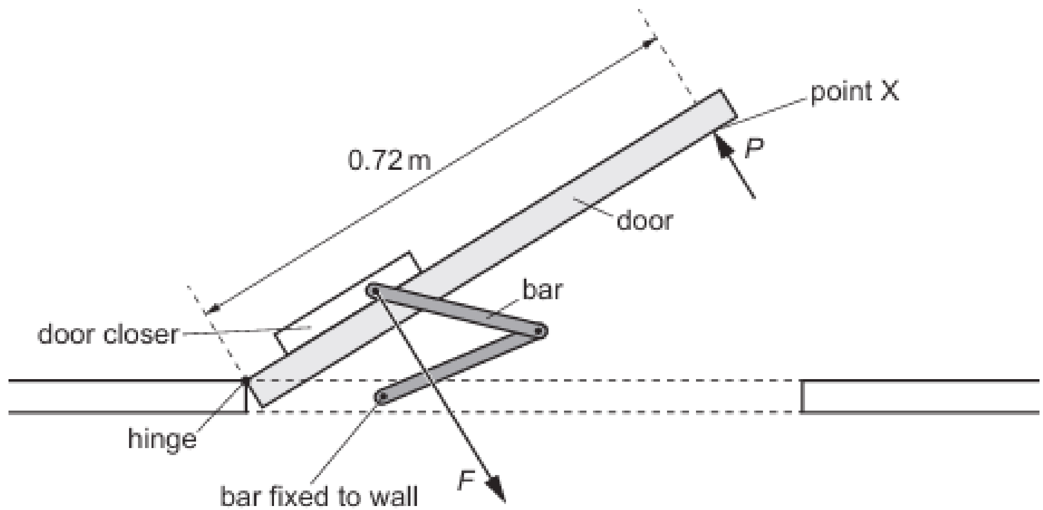


Fig. 3.1

When the door opens and closes, the hinge acts as a pivot.

A girl opens the door by exerting a force *P* at point X.

Force *P* is perpendicular to the surface of the door.

(a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge. The force *P* is 25 N.

(i) Calculate the moment of force *P* about the hinge.

moment of force = ... [2]

(ii) The door rotates about the hinge by 90°. The circumference of a circle of radius 0.72 m is 4.5 m.

Calculate the work done on the door by force *P*.

work done = ... [2]

(b) As the door opens, there is a force *F* on the door in the direction shown in Fig. 3.1.

Although force *F* is larger than force *P*, the door rotates about the hinge.

Explain why.

....

....

....

.... [2]

[Total: 6]

7. O/N/20/22/Q1

A glass beaker has a mass of 50 g. A liquid of density 1.8 g / cm^3 is poured into the beaker until it reaches the 200 cm^3 mark.

(a) Calculate the total mass of the beaker and its contents.

mass = ... [3]

(b) The centre of mass of a metre rule is at the 50 cm mark.

(i) State what is meant by *centre of mass*.

....

.... [1]

(ii) The metre rule is placed on a pivot. The tip of the pivot is under the 80 cm mark on the rule.

The beaker with its contents is then placed at different positions along the rule until the rule is balanced.

Fig. 1.1 shows the arrangement with the rule balanced.

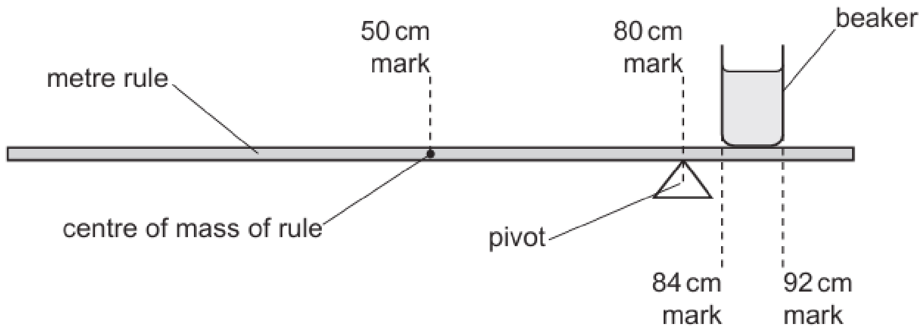


Fig. 1.1

One side of the beaker is at the 84 cm mark and the other side is at the 92 cm mark.

Calculate the mass of the rule.

mass = ... [3]

[Total: 7]

8. M/J/20/21/Q2

A student performs an experiment to mark the centre of mass C on a thin piece of card. There are two holes in the card.

Fig. 2.1 shows the card and two lines that the student draws on the card.

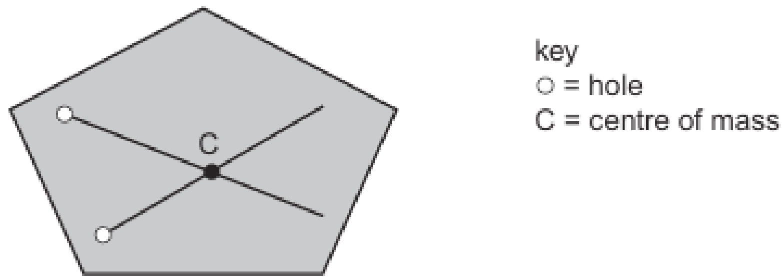


Fig. 2.1

(a) Describe a method used to draw these two lines in their correct positions on the card.

Make clear what extra apparatus is needed. You may draw a diagram, if you wish.

....

....

....

....

....

....

.... [3]

(b) The student holds the card loosely between her fingers. The card is vertical, resting with its lower edge on a bench as shown in Fig. 2.2.

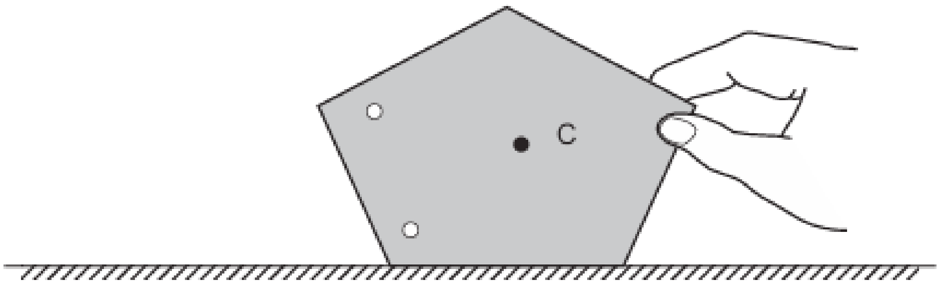


Fig. 2.2

The card is tilted slightly, as shown in Fig. 2.3, and then released.

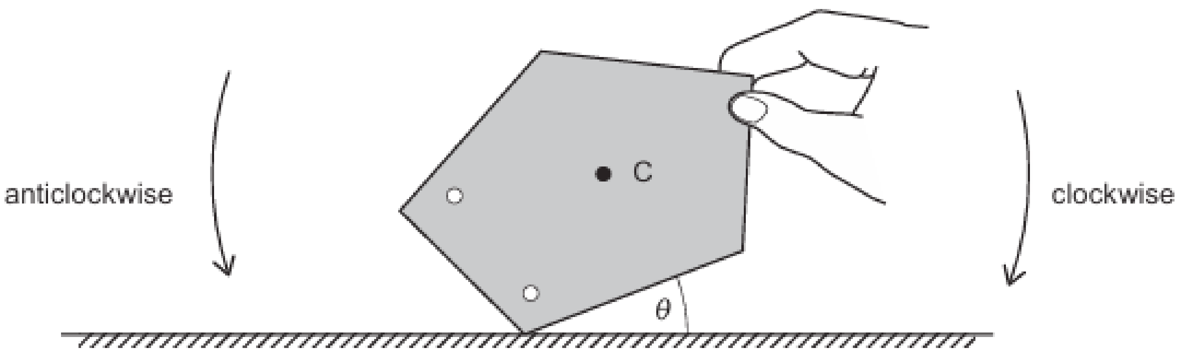


Fig. 2.3

When angle θ is small, the card falls clockwise, back to the position shown in Fig. 2.2.

(i) Explain why the card falls anticlockwise when θ is large.

....

....

....

.... [2]

(ii) State one change to the card that makes it more stable.

....

.... [1]

[Total: 6]

9. O/N/18/21/Q8

A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

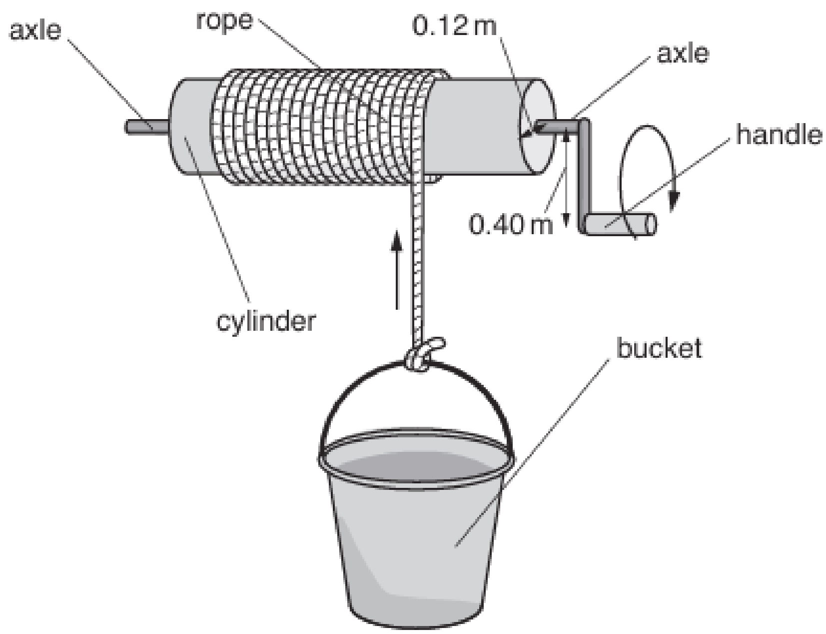


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

(a) Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of ....

in the body. It resists a change in the state of .... of

the body. [2]

(b) When the bucket is full, it contains $2.4 \times 10^{-2}\,m^3$ of water.

The gravitational field strength *g* is equal to 10 N/kg.

(i) Explain what is meant by a *gravitational field*.

....

.... [1]

(ii) The density of water is $1000\,kg/m^3$.

Determine the total weight of the bucket and the water.

weight = ... [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = ... [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = ... [1]

10. M/J/17/22/Q3

Fig. 3.1 shows the brake pedal of a car which is connected to a brake cylinder.

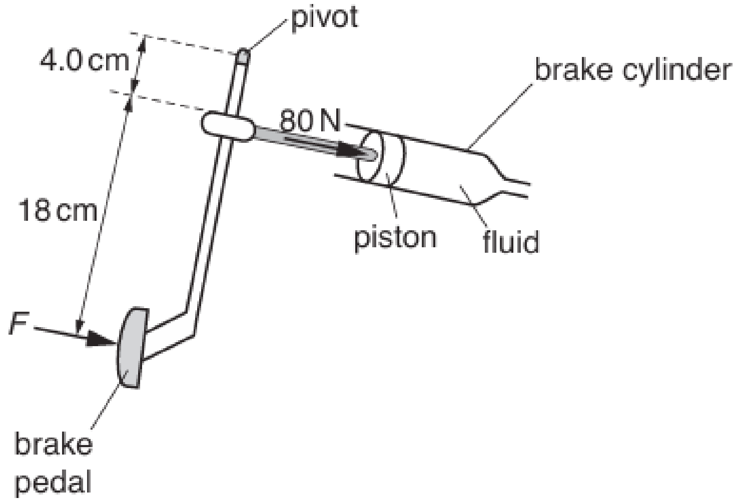


Fig. 3.1 (not to scale)

The brake is pressed with a force *F*. This force produces a moment about the pivot.

Pressing the brake causes a force of 80 N to act on the piston.

(a) Define the term *moment*.

....

....[2]

(b) Calculate the force *F* applied to the brake pedal.

F = ...[2]

11. O/N/16/22/Q3

When the lid of a freezer is opened, it pivots about the hinge at the back of the freezer. The handle is at the front.

Fig. 3.1 is a side view of the freezer.

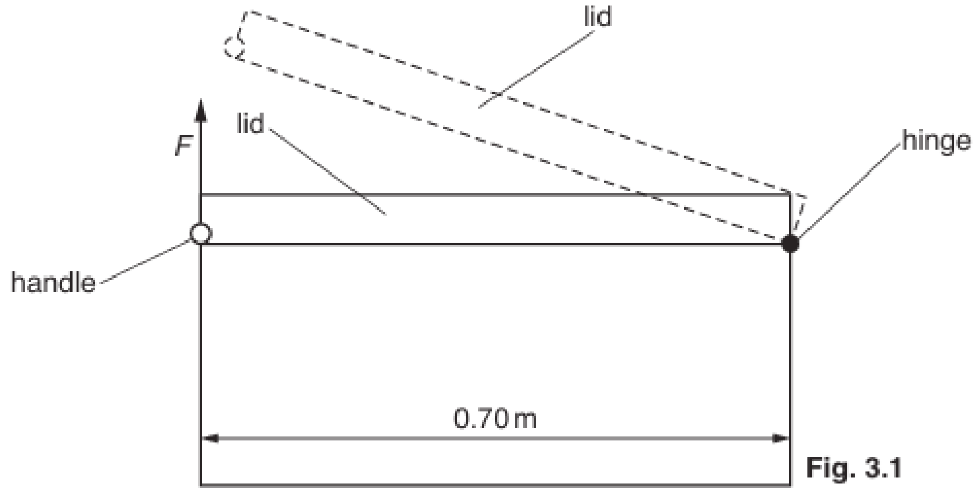


Fig. 3.1

The handle is 0.70 m from the hinge. The lid has a mass of 2.0 kg.

(a) The gravitational field strength g is 10 N / kg.

(i) Calculate the weight of the lid.

weight = ...[1]

(ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment = ...[2]

2. The moment required to open the lid is greater than the value calculated in **(a)(ii)1.**

Suggest one reason for this.

....

....[1]

(b) The lid is closed. To open the lid, a force F is applied to the handle as shown in Fig. 3.1.

The direction of F is vertically upwards and F is the smallest possible force that opens the lid.

A force on the handle in any other direction must be larger than F in order to open the lid. Explain why.

....

....[1]

12. O/N/16/21/Q2

Fig. 2.1 shows a bed that folds away against a wall during the day.

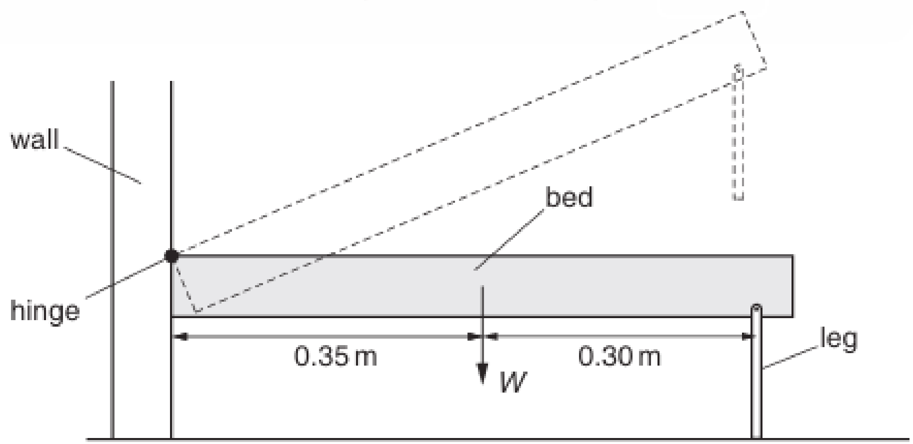


Fig. 2.1 (not to scale)

When it is horizontal, the bed is supported on one side by a hinge and on the other side by two legs. The weight W of the bed acts through its centre of mass, which is at a horizontal distance of 0.35 m from the hinge.

(a) The mass of the bed is 26 kg. The gravitational field strength g is 10 N/kg.

Calculate the weight W of the bed.

W = ...[1]

(b) (i) State the principle of moments.

....

....

....[2]

(ii) Determine the size of the total upward force exerted on the bed by the two legs when the bed is horizontal.

total force = ...[3]

13. O/N/15/21/Q4

Fig. 4.1 shows a screwdriver of mass 64 g resting in equilibrium on a pivot.

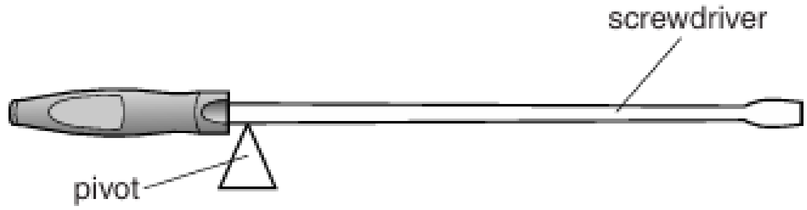


Fig. 4.1

(a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]

(b) The gravitational field strength is 10 N / kg.

(i) Calculate the weight of the screwdriver.

weight = ... [1]

(ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

(c) State two conditions that apply when an object is in equilibrium.

1. ....

....

2. ....

....

[2]

14. M/J/15/21/Q2

Fig. 2.1 shows a student doing a press-up. A total force *F* acts upwards on his hands. There is also a force upwards on his toes.

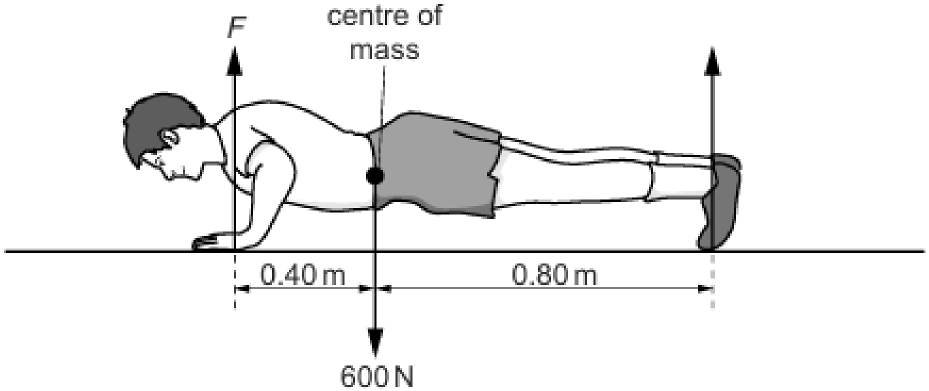


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

(a) **(i)** Explain why the student does work as his body rises from the ground.

....

....[1]

(ii) State the form of energy that the student uses to do this work.

....[1]

(b) At the position shown in Fig. 2.1, the student is stationary.
The weight of the student causes a moment about his toes.

(i) Calculate the moment of the weight of the student about his toes.

moment = ...[1]

(ii) Calculate the value of the force *F*.

F = ...[2]

15. O/N/14/22/Q2
In hospitals, doctors and nurses operate taps with their elbows in order to avoid contamination.

Fig. 2.1 shows a hospital tap with a long handle.

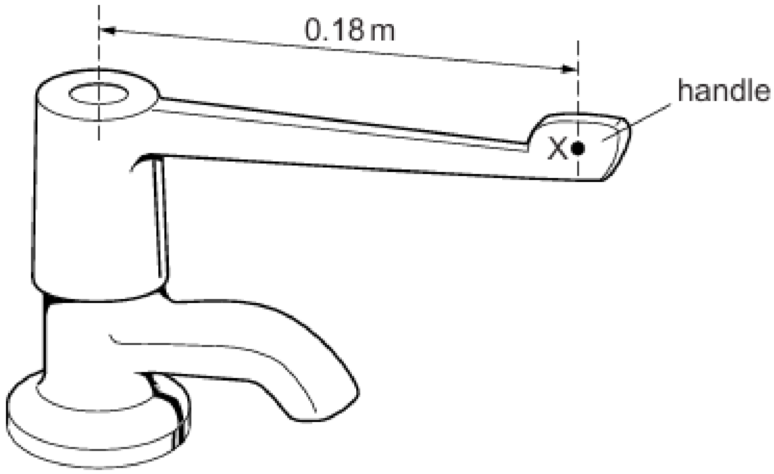


Fig. 2.1

(a) A nurse applies a force of 2.5 N at a point X on the handle, 0.18 m from the axis of the tap.

(i) Calculate the maximum moment about the axis that this force can produce.

moment = ...[2]

(ii) The moment produced by the nurse is less than this maximum value.

Suggest one reason why this is so.

....

....[1]

(b) State how the force needed to operate the tap is affected by the length of the handle.

....

....[1]

16. O/N/13/22/Q9

(a) State what is meant by the *moment of a force.*

....

....[2]

(b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

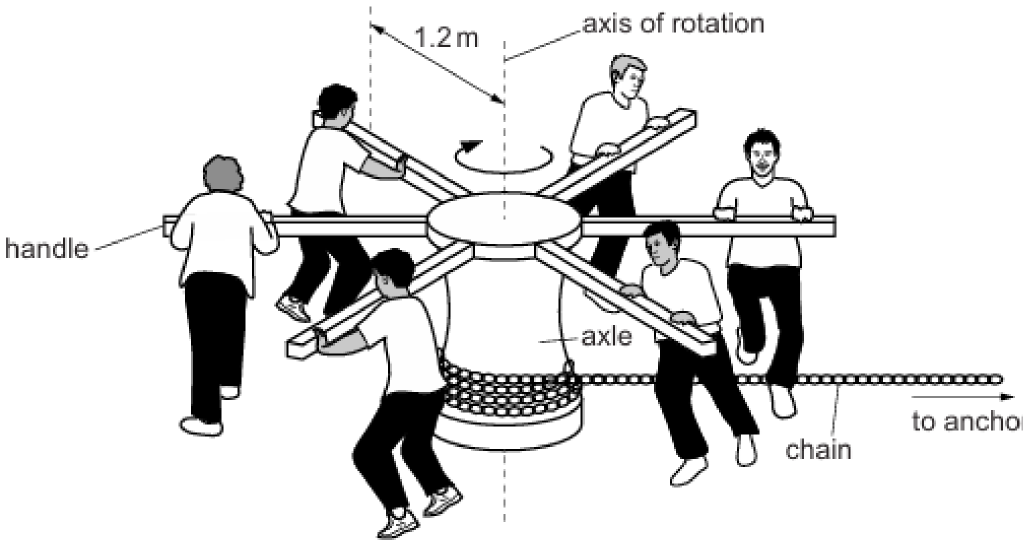


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength *g* is 10 N / kg.

(i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment = ...[2]

2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

energy = ...[3]

(ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

....

....

....[2]

(iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

....

....[2]

(c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

....

....

....

....

....

....[4]

17. O/N/13/21/Q2

A girl of weight 550 N is playing on a see-saw with her brother. Fig. 2.1 shows her brother of weight W sitting 1.1 m to the right of the balance point.

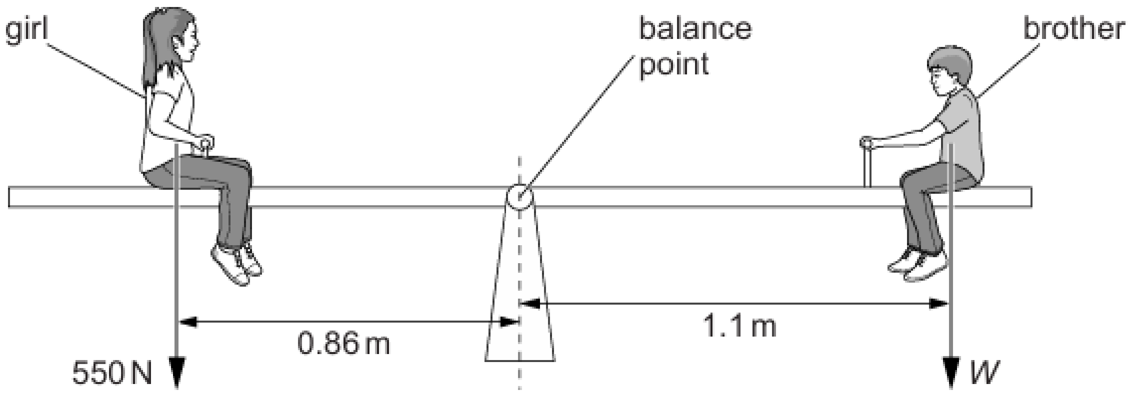


Fig. 2.1 (not to scale)

The see-saw is balanced when the girl sits 0.86 m to the left of the balance point.

(a) Calculate W.

W = ...[2]

(b) The girl and her brother slide equal distances along the see-saw away from each other.

Describe and explain what happens.

....

....

....

....[3]

18. M/J/13/22/Q2

(a) Explain what is meant by the *moment of a force*.

....

....

.... [2]

(b) Fig. 2.1 shows a system for raising a heavy piece of metal into a vertical position.

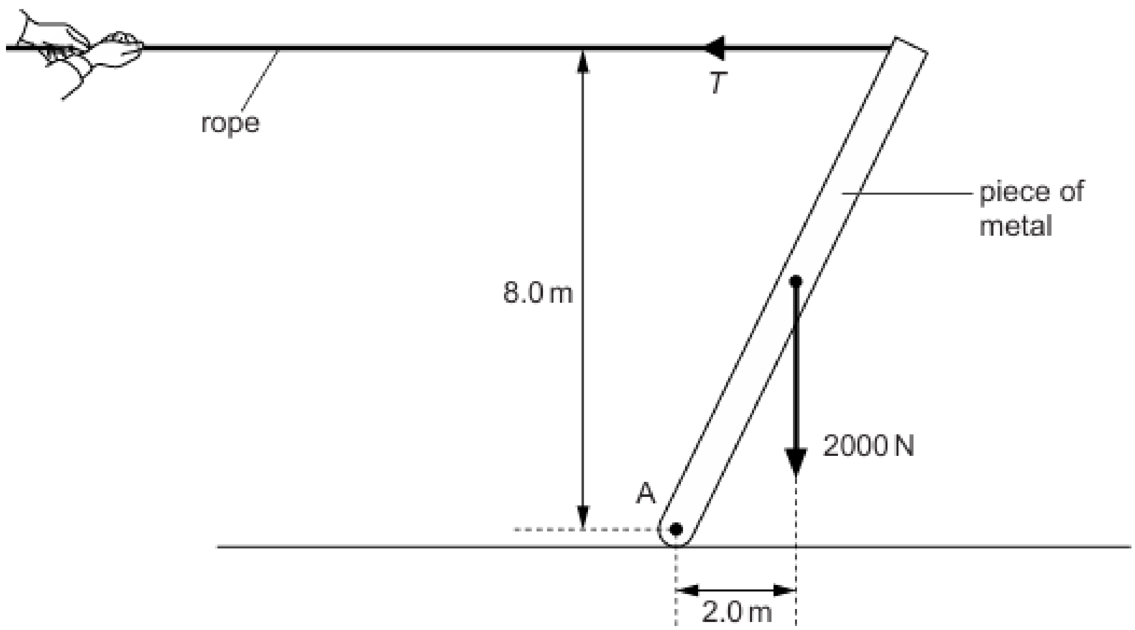


Fig. 2.1 (not to scale)

A man pulls on the rope with a horizontal force T. The piece of metal has a weight of 2000 N and is freely pivoted at A. The system is in equilibrium.

(i) By taking moments about A, calculate T.

T = ... [2]

(ii) The force T and the force that the rope exerts on the man are related by Newton's third law. State the relationship between these forces.

....

....

.... [2]

19. M/J/13/21/Q2

Fig. 2.1 shows apparatus used to investigate the turning effect of a force.

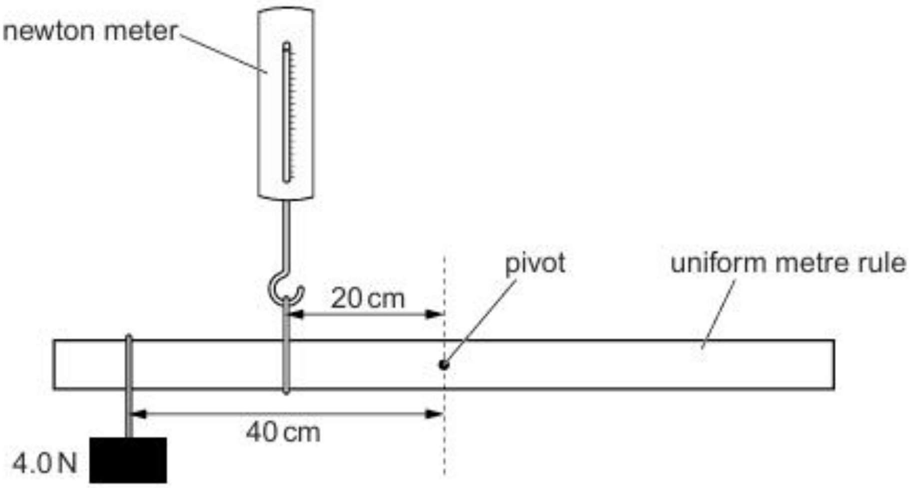


Fig. 2.1

The uniform metre rule is freely pivoted at its centre.

The newton meter is 20 cm from the pivot and a 4.0 N weight is 40 cm from the pivot.

The metre rule is in equilibrium.

(a) State the principle of moments for a body in equilibrium.

....

....[1]

(b) Calculate the reading on the newton meter.

reading = ...[2]

(c) The weight of the metre rule is 2.2 N.

Calculate the size of the force exerted on the metre rule by the pivot.

force = ...[2]

20. O/N/12/22/Q1

(a) Describe an experiment to verify *the principle of moments*. You may include a diagram in your answer.

....

....

....

....

....[4]

(b) Fig. 1.1 shows a spanner tightening a nut.

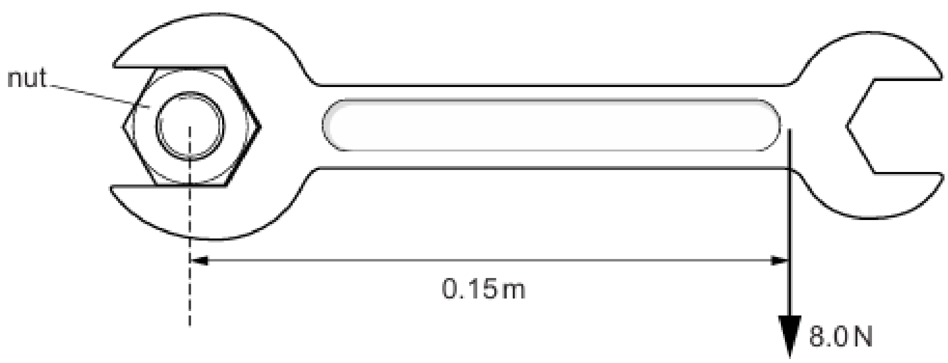


Fig. 1.1

A force of 8.0 N is applied to the spanner at a perpendicular distance of 0.15 m from the centre of the nut.

Calculate the moment of the force acting on the nut.

moment = ...[2]

21. O/N/11/21/Q11

A crankshaft is a shaped metal bar that is part of a car engine. It is free to rotate about an axis, as shown in Fig. 11.1.

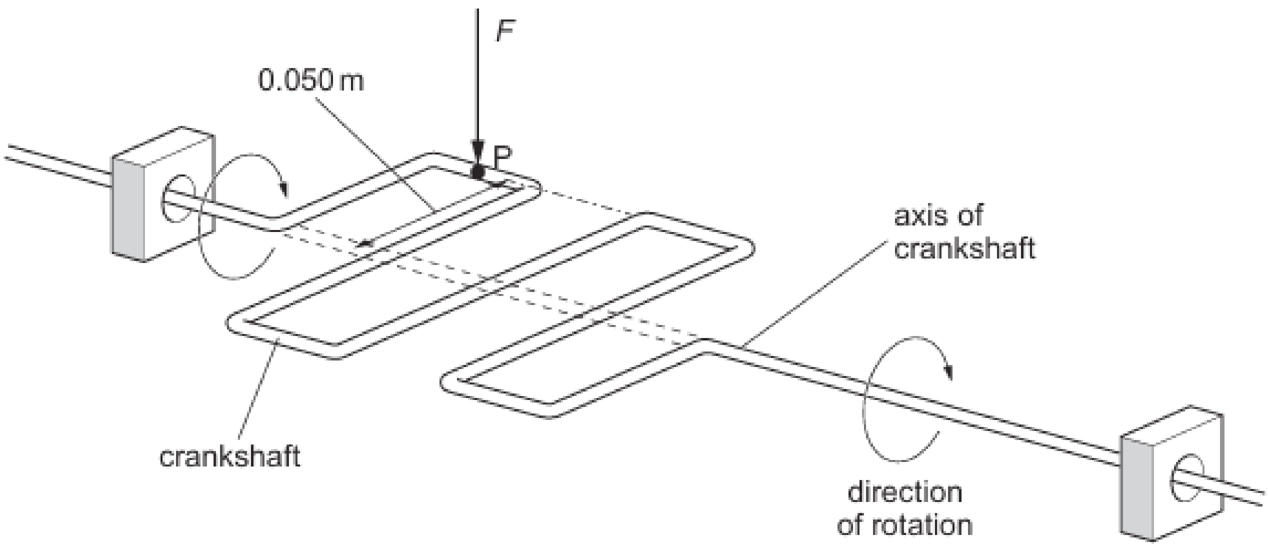


Fig. 11.1

When the crankshaft is horizontal, a vertical force F of 8200 N acts downwards on the crankshaf at P. This causes the crankshaft to rotate. The distance between P and the axis of the crankshaft is 0.050 m.

(a) **(i)** State what is meant by the *moment* of a force.

....

....[2]

(ii) The crankshaft is horizontal. Calculate the moment of F about the axis of the crankshaft.

moment = ...[2]

(iii) The size and direction of the vertical force F, acting on the crankshaft at P, remain constant. Explain why the moment of F decreases as the crankshaft rotates through a small angle.

....

....[1]

22. O/N/10/22/Q3

Fig. 3.1 shows a firefighter of total weight 840 N in equilibrium at the top of a ladder that is pivoted at point P.

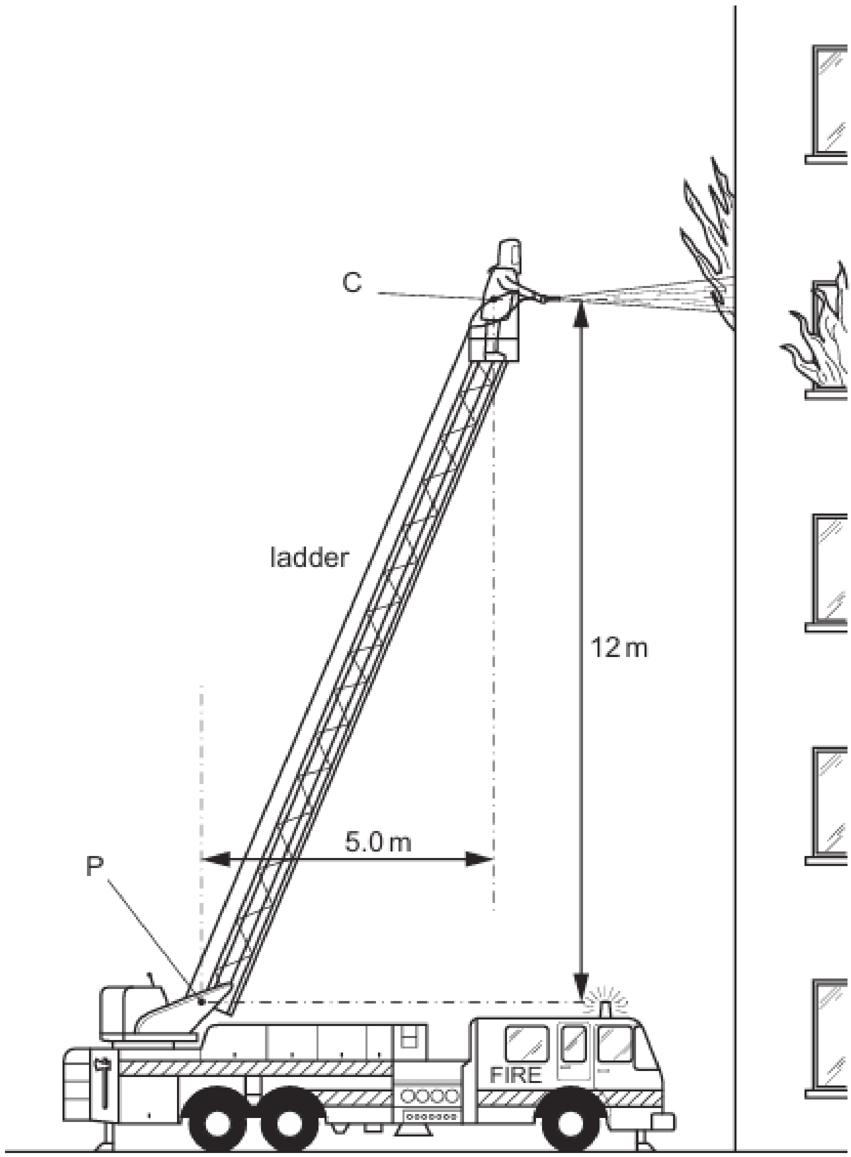


Fig. 3.1

The ladder leans towards a burning building at an angle such that the centre of gravity C of the firefighter is 12 m above and 5.0 m to the right of P. The firefighter holds a hose that directs a high-speed jet of water horizontally into a burning building.

(a) (i) Calculate the moment *M* of the firefighter's weight about P.

moment = ... [2]

(ii) The jet of water causes a horizontal force R on the firefighter that acts towards the left, through C. This opposes the turning effect of his weight. Calculate the size of R that, on its own, ensures that M is exactly cancelled.

force = ...[1]

(iii) Suggest a third force that has a turning effect about P on the ladder.

....

....[1]

23. O/N/09/Q7

Fig. 7.1 shows a rectangular concrete slab of weight 18000 N. It rests on a brick wall and is the roof of a bus shelter. The concrete slab is 3.0 m wide.

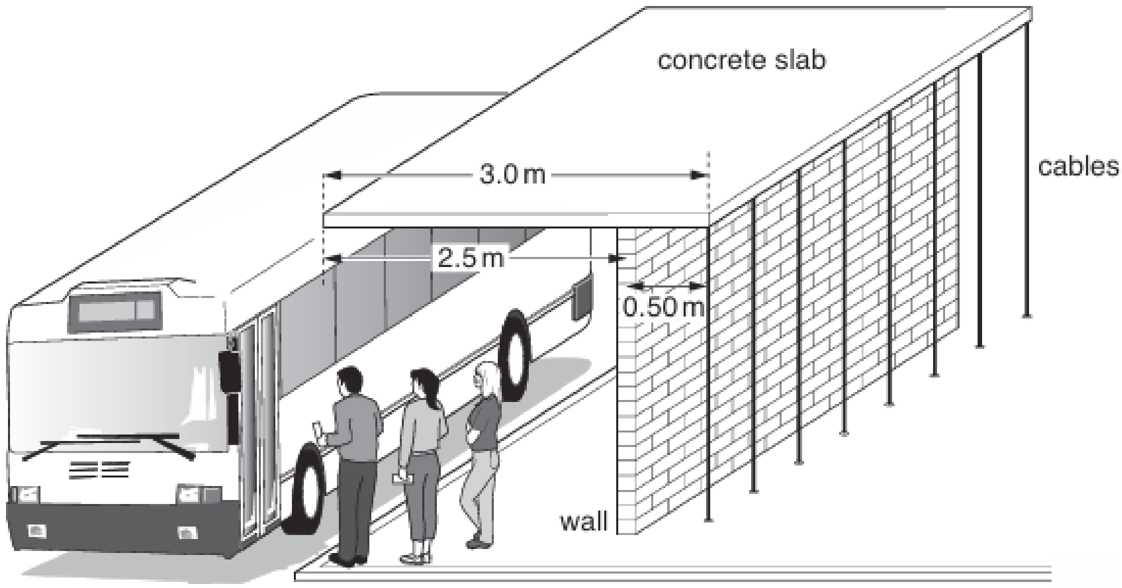


Fig. 7.1

The wall is 2.5 m from the front of the concrete slab and 0.50 m from the back. The cables behind the shelter pull downwards and stop the slab toppling forwards.

(a) The concrete slab is of uniform thickness and density. Determine the perpendicular distance between the wall and the centre of mass of the slab.

distance = ... [1]

(b) **(i)** State the principle of moments.

....

....

.... [2]

(ii) Calculate the total downward force exerted by the cables on the slab.

force = ... [2]

ANSWERS

1.

Question	Answer	Marks
3(a)(i)	it / lamina is in the gravitational field (of Earth)	**B1**
3(a)(ii)	(F =) mg **or** 0.050 × 9.8 **or** 50 × 9.8 **or** 490	**C1**
	0.49 (N)	**A1**
3(b)(i)	(M =) $F \times x_{\perp r}$ **or** 0.49 × 8.0 **or** 0.49 × 0.080	**C1**
	0.039 (N m)	**A1**
3(b)(ii)	it / lamina moves clockwise (about the pivot)	**B1**
	it speeds up **and** then slows down **or** it oscillates (with decreasing amplitude) **or** overshoots	**B1**
	(stops with) G vertically below H	**B1**

2.

Question	Answer	Marks
1(a)(i)	longer	**B1**
	thinner **or** smaller (cross-sectional) area	**B1**
1(a)(ii)	elastic potential **or** strain (energy)	**B1**
1(b)(i)	sum of / total / net / resultant moment is zero	**M1**
	when (the object is) in equilibrium	**A1**
1(b)(ii)	distance of rubber band to pivot is smaller than the distance of the mass (to pivot)	**B1**
1(b)(iii)	4 × 18 = F × 2	**C1**
	36 N	**A1**
1(b)(iv)	greater **and** the weight (of XY) contributes to the (clockwise) moment/turning effect **or** the moment of the rubber band force is larger	**B1**

3.

Question	Answer	Marks
1(a)	(p =) F / A or 2000 / (2 × 0.040) **or** 50 000 (P–a)	**C1**
	25 000 Pa	**A1**
1(b)(i)	symmetrical left-to-right **or** symmetrical front-to-back **or** bench is stable	**B1**
	because centre of mass is in middle (horizontally) **or** centre of mass is within the base	**B1**
1(b)(ii)	(moment =) Fx **or** Wy **or** 2000 × (1.1 – 0.25) **or** 2000 × 0.85 clockwise moment = anticlockwise moment **or** $Fx = Wy$ **or**	**C1**
	Fx = 2000 × (1.1 – 0.25) **or** Fx = 2000 × 0.85 **or** (F =)2000 × 0.85 / 0.25	**C1**
	6800 N	**A1**

4.

Question	Answer	Marks
8(a)	$F \times$ distance / displacement	**M1**
	perpendicular distance to line of action of force shown on Fig. 7.1	**A1**
8(b)(i)	for an object in equilibrium	**B1**
	(sum of) clockwise moments is equal to (sum of) anticlockwise moments **or** no resultant moment	**B1**
8(b)(ii)	equipment shown in diagram (e.g. rule, two loads, pivot)	**B1**
	what is done to achieve balance (e.g. balance rule on pivot, one load each side of pivot)	**B1**
	how principle is verified (e.g. calculate two opposite moments and find they are equal)	**B1**
8(c)(i)	80 N	**B1**
8(c)(ii)	(point) where the (whole) weight of an object acts / seems to act **or** point where the object balances	**B1**

Question	Answer	Marks
8(c)(iii)	mass not uniformly distributed along length **or** more mass at the one end	**B1**
8(c)(iv)	(Γ =) 80×0.90	**C1**
	72 N m	**A1**
8(c)(v)	moment / 0.54 **or** 209 (N) **or** 41.1 / 0.54	**C1**
	76 N	**A1**
	downwards	**B1**

5.

Question	Answer	Marks
3(a)	(mass =) meter reading / g meter reading / gravitational field (strength meter reading / acceleration due to gravity	**B1**
3(b)(i)	(sum of) clockwise moments = (sum of) anticlockwise moments (about any point)	**B1**

Question	Answer	Marks
3(b)(ii)	any attempt at using moments, e.g. $3 \times 30 = F \times 20$	**C1**
	4.5 N	**A1**
3(c)	force upwards (on brick due to water) **or** force due to water (on brick) **or** force downwards (on brick) is reduced / counteracted (by water) **or** force exerted by brick (on rule) reduced **or** force / tension in string reduced	**B1**
	(force caused by) pressure (of water) **or** reduced anticlockwise moment	**B1**

6.

Question	Answer	Marks
3(a)(i)	(Γ =) $Fx_{\perp r}$ **or** 25×0.72	**C1**
	18 N m	**A1**
3(a)(ii)	(WD =) $Fx_{\parallel}$ **or** $F\pi r / 2$ or $25 \times 4.5 / 4$ **or** 25×1.13	**B1**
	28 J	**B1**
3(b)	moment due to force P is greater	**B1**
	distance of X to hinge is greater than distance from where force F acts	**B1**

7.

Question	Answer	Marks
1(a)	$(m =)\ \rho V$ **or** 1.8×200 **or** 360	**C1**
	$(1.8 \times 200) + 50$ **or** $360 + 50$ **or** 360 **or** candidate's mass + 50	**C1**
	410 g **or** 0.41 kg	**A1**
1(b)(i)	(place / region) where the mass appears to be concentrated **or** where the object balances	**B1**
1(b)(ii)	88 (cm) seen **or** $(84 + 92) / 2$ seen **or** 30 (cm) seen **or** 8(.0 cm) seen **or** 164 (g)	**C1**
	$m_1gx_1 = m_2gx_2$ **or** $F_1x_1 = F_2x_2$ **or** $W_1x_1 = W_2x_2$ **or** $(m_1 =)\ m_2x_2 / x_1$ **or** $410 \times (88 - 80) / 30$ **or** 164 (g)	**C1**
	110 g **or** 0.11 kg	**A1**

8.

Question	Answer	Marks
2(a)	Hang card from pin / nail though hole **and** wait until stationary	**B1**
	using plumb line from hole mark line (on card)	**B1**
	repeat with another hole	**B1**
2(b)(i)	centre of mass / C falls outside base	**B1**
	weight creates moment / turning effect	**B1**
2(b)(ii)	lower centre of gravity **or** wider / longer base	**C1**

9.

(a) substance **or** matter **B1**
rest **or** motion **B1**
(b)(i) (a region of space) where a mass experiences a force (due to gravitational attraction) **B1**
(b)(ii) $(m =)\ \rho V$ **or** $1000 \times 2.4 \times 10^{-2}$ **or** 24 (kg) **C1**
or **C1**
250 N
25 (kg) 240 (N) **A1**
(c)(i) $(\Gamma =)\ Fx$ **or** 250×0.12 **C1**
30 N m **A1**
(c)(ii) 75 N **B1**

10.

(a) force × distance **C1**
ignore force into distance

force × **perpendicular** distance (from line of action to point / pivot) **A1**

(b) any moment calculation seen, e.g. $F \times 22 = 80 \times 4$ **C1**

15 N **A1**

11.

(a) **(i)** 20 N B1

(ii) **1.** (Γ =)Fd **or** 20 × 0.35 **or** 20 × 0.70 **or** 14 C1

7.0 N m A1

2. friction (at hinge / seal) **or** air resistance **or** to cause an initial acceleration B1

(b) (for other directions) **perpendicular** distance is less B1

12.

(a) 260 N B1

(b) **(i)** for a body in equilibrium B1

(total) clockwise moment = (total) anticlockwise moment B1

(ii) $F_1d_1 = F_2d_2$ **or** 260 × 0.35 **or** 91 **or** F × 0.65 C1

260 × 0.35 = F × 0.65 **or** 260 × 0.35/0.65 **or** 91 = F × 0.65 **or** 91/0.65 C1

140 N A1

13.

(a) (point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (±1 mm)) B1

(b) **(i)** 0.64 N B1

(ii) arrow W vertically downwards through candidate's C or pivot B1

(c) no resultant force **or** upward force = downward force **or** force left = force right B1

no resultant moment (of force) **or** clockwise moment = anticlockwise moment B1

14.

(a) **(i)** force moves through a distance (in same direction) B1

(ii) chemical (potential) energy B1

(b) **(i)** 480 Nm B1

(ii) attempt to apply moments with two forces and distances C1

400N A1

15.

(a) **(i)** Fd **or** 2.5 × 0.18 C1

0.45 N m A1

(ii) force not applied at right angles to the tap B1

(b) long(er) distance needs small(er) force (for same moment) **or** inversely related/proportional B1

16.

(a) force × distance **or** $F \times d$ with F and d defined **or** $F \times d_{perp}$ — C1
force × perpendicular distance **or** $F \times d_{perp}$ with F and d_{perp} defined — A1

(b) (i) 1. 6 × 750 × 1.2 **or** 750 × 1.2 **or** 900 — C1
5400 N m — A1

2. *mgh* **or** 350 × 10 × 160 **or** 350 × 10 × 1.6 — C1
350 × 10 × 1.6 **or** 5.6 × 105 — C1
5600J — A1

(ii) friction at axle/boat **or** drag due to water
or chain lifted also — B1
heat produced (**ignore** in sailors) **or** work done against friction/drag
or work done raising chain — B1

(iii) same amount of work done **or** $P = E/t$ **or** $P = \text{WD}/t$ — B1
in less time **or** power inversely proportional to time (**ignore** faster rate) — B1

(c) clear/labelled diagram with ruler, fulcrum, at least two weights — B1
any three of the following points made in words:
balance ruler (on its own)
place weights on ruler so it balances
clockwise and anticlockwise moments equal **or** net moment zero
repeat (apply similar principles to other possible methods) — B3

17.

(a) $F_1x_1 = F_2x_2$ **or** 550 × (0.86 or 86)/(1.1 or 110) — C1
430 N — A1

(b) **both** moments increase — C1
girl's moment increases more **or** girl's moment > brother's
or anticlockwise moment greater — A1
see-saw tips down on girl's side — B1

18.

(a) force × distance — M1
perpendicular distance — A1

(b) (i) $T \times 8$ or 2000 × 2 seen — C1
500 N — A1

(ii) (two forces) equal (in magnitude) — B1
(two forces) opposite (in direction) — B1

19.

(a) total/resultant **moment** zero **or** (sum of) clockwise = anticlockwise **moment** — B1

(b) F_1d_1 **or** F_2d_2 seen in any form — C1
8(.0) N — A1

(c) 4 + 1.2 or 5.2 seen — C1
2.8 N ecf **(b)** i.e. accept 5.2 – **(b)** or **(b)** – 5.2 — A1

20.

(a) appropriate apparatus e.g. ruler, weights, fulcrum B1
action e.g. balance weights on each side B1
one of: force/mass × distance **or** calculate moment B1
vary **or** repeat B1

(b) *F* × *d* **or** 8.0 × 0.15 C1
1.2 N m (**not** J) A1

21.

(a) **(i)** force × distance C1
force × perpendicular distance (from the axis) A1

(ii) 8200 × 0.05 C1
410 N m A1

(iii) (perpendicular) distance reduced/force not perpendicular/only a component of the force is perpendicular B1

22.

(a) **(i)** (*M* =) force × **perpendicular** distance **or** 840 × 5
(formula mark can be scored if not given in **3(a)(ii)**) C1
4200 N m A1

(ii) 350 N **or** **(a)(i)**/12 **and** calculated B1

(iii) weight of ladder/hose **or** friction at P/pivot/axle
(**not** air resistance; **ign**. friction) B1

23.

(a) 1.(0) m B1

(b) **(i)** (for an object in) equilibrium/balance B1
$W_1x = W_2y$ (clear) **or** anticlockwise moment/torque/turning force = clockwise moment/torque/turning force B1

(ii) 18 000 × 1. 0 = T × 0.5 C1
36 000 N A1